EUCLIDEAN MINIMALITY OF MARKOV REPRESENTATIONS ON THIRTEEN FAREY ROWS

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Abstract. The Markov number attached to a reduced Farey fraction is canonically a sum of two coprime squares, and the subtractive Euclidean algorithm reduces this pair in exactly as many steps as the numerator plus the denominator minus two. We prove that no other primitive representation of the same integer as a sum of two squares has a shorter Euclidean algorithm when the numerator is at most thirteen, with no bound on the denominator. A parity-order theorem for the Fibonacci-normalized norm reduces infinite families of Euclidean paths to exact comparisons, complete Gaussian factorizations settle finitely many labels, and the remaining finite assertions are verified by computer in exact arithmetic. Minimality on every row would imply the Frobenius–Markov uniqueness conjecture. We conjecture that it holds, and state a second conjecture, on iterated palindromic closure and the Thue–Morse morphism, that also implies uniqueness and is verified for every Markov number up to 10^{12}.

1. A representation problem attached to Markov numbers

The Markov equation

$$x^2 + y^2 + z^2 = 3xyz$$

has a tree of positive integer solutions generated from $(1, 1, 1)$ by permutations and the replacement $(x, y, z) \mapsto (x, y, 3xy - z)$. The Frobenius–Markov uniqueness conjecture asks whether the largest entry determines a solution up to permutation. A useful approach is to attach an arithmetic optimization problem to each occurrence of that entry in the tree. If two occurrences are both optimal, a known ordering theorem can distinguish them.

We use the Farey indexing $m_{P/Q}$ of Markov numbers, with $0 < P < Q$ and $\gcd(P, Q) = 1$, as in [1] and [14]. An occurrence includes its reduced fraction; its label is the integer $m_{P/Q}$. Keeping these notions separate is essential in a uniqueness problem. The first labels in our convention are $m_{1/2} = 5$, $m_{1/3} = 13$, and $m_{2/3} = 29$.

Every such occurrence supplies a primitive positive representation $m_{P/Q} = r^2 + s^2$. The subtractive Euclidean algorithm gives this pair an integer cost $\ell(r, s)$: count single subtractions until $(1, 1)$ is reached. The cost can be read in two equivalent ways, and the paper uses both. If $s/r = [q_0; \dots, q_j]$ is the canonical finite continued fraction, with last quotient at least two, the cost is the sum of the partial quotients minus one, $q_0 + \cdots + q_j - 1$. In matrix form,

$$\begin{pmatrix} s \\ r \end{pmatrix} = P_{q_0} \cdots P_{q_{j-1}} P_{q_j - 1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \qquad P_q = \begin{pmatrix} q & 1 \\ 1 & 0 \end{pmatrix},$$

and the cost is the sum of the indices of the factors; Lemma 2.1 proves that the two readings agree. The occurrence pair has cost $P + Q - 2$; this follows from the continued fraction of Çanakçı and Schiffler [6], as we recall in Section 2. Our question is whether any

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other primitive representation of the same integer can have smaller cost. The competitors include all partial quotients and need not arise from another Markov occurrence.

Theorem 1.1 (Thirteen Farey rows)**.** *Let $0 < P < Q$ be coprime integers with $P \leqslant 13$. For every primitive positive representation*

$$m_{P/Q} = r^2 + s^2, \qquad 0 < r < s,$$

the subtractive Euclidean cost satisfies

$$\ell(r, s) \geqslant P + Q - 2,$$

with equality for the occurrence-induced representation.

There is no upper bound on Q or on the label. The minimum is weak: the theorem permits other representations of the same cost. For the 248 occurrences of the finite windows of Section 3, the occurrence pair is moreover the only representation of minimum cost. Since equal labels with equal costs come from the same fraction (Lemma 2.5), weak minimality separates occurrences.

Corollary 1.2. *If P/Q and P'/Q' are reduced fractions of $(0, 1)$ whose numerators are at most thirteen, then $m_{P/Q} = m_{P'/Q'}$ implies $P/Q = P'/Q'$.*

Corollary 1.2 is not new in strength. Lee, Li, Rabideau and Schiffler show that two distinct fractions with equal labels have a slope $(P - P')/(Q - Q')$ strictly between $-5/4$ and $-8/7$, and their numerators then differ by at least six [16, Corollary 1.5]; this gives the injectivity on the first six rows, and on the first thirteen rows it leaves sixteen one-parameter families of candidate pairs, which their fixed-row limits [16, Lemma 6.4 and (6.4)] (for the first row, the exact value $m_{1/Q} = F_{2Q+1}$) and a few explicit comparisons of labels exclude. What Corollary 1.2 adds is its route: it comes from a minimality principle, which makes sense on every row and would give the uniqueness conjecture if it held on every row. The new content of the paper is Theorem 1.1 itself, a statement about every primitive representation of the label, Markov or not, and the method that proves it.

1.1. **The mechanism of the proof.** Two descriptions of a primitive pair organize the argument. Gaussian factorization enumerates every representation of one fixed integer. Inverse Euclidean steps organize every pair of one fixed cost. The first description treats a finite set of labels completely; the second controls the unbounded part of each row. Figure 1 shows how the two descriptions divide the thirteen rows between them.

At cost c, the largest norm is the Fibonacci value F_{2c+3}. Every other path departs from the Fibonacci path at some steps, called defects. Record the lengths of the Fibonacci runs between successive defects. The main general tool proved here, Theorem 5.1, is that, on fixing their parities, the full normalized norm N/F_{2c+3} is strictly monotone in each run length, with an explicit direction. This holds for every positive number of defects, not only for the strata used in the thirteen-row theorem. It reduces each infinite box of run lengths to two exact corner values in $\mathbb{Q}(\sqrt{5})$.

An exact expansion of the Cohn trace, in the eigenbasis of the first Cohn matrix, places the source labels of each row in narrow normalized intervals around an explicit constant, and we show that every shorter path misses the interval belonging to its row. The covers used at that step admit unbounded run lengths: their role is to prove uniform inequalities, not to extrapolate from a finite search. What remains after them is a finite list of labels, for which the complete two-square fiber is determined by Gaussian factorization. The finitely many assertions that these two steps leave to a computer are collected in one proposition, Proposition 6.1; everything else is proved in the text.

Section 2 gives the two readings of the cost, recalls the occurrence cost identity and proves the restricted injectivity reduction. Section 3 describes the complete finite fibers

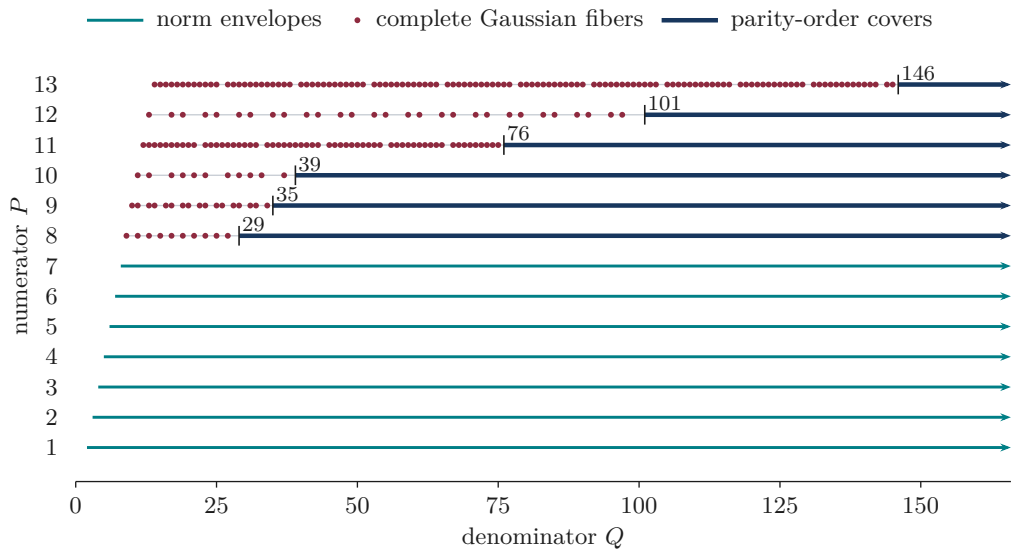


FIGURE 1. How the proof covers the thirteen rows. Each horizontal line is a Farey row of numerator P, drawn against the denominator Q. The rows $P \leqslant 7$ are settled by the norm envelopes of Section 4 (teal). In the rows $8 \leqslant P \leqslant 13$ the dots are the 248 reduced fractions of the finite windows, settled by complete Gaussian fibers in Section 3 (wine); from the junction denominator Q_P on, marked by a tick, the unbounded tail is settled by the parity-order covers of Section 5 and the comparisons of Section 6 (navy). The arrows indicate that the rows continue without bound.

and locates the occurrence pair in them, by one exact Gaussian division per step of the Farey tree (Proposition 3.8). Section 4 develops the path envelopes and proves the first seven rows. Section 5 establishes the general parity order and the construction of the covers. Section 6 states the finite verifications, draws their consequences, joins the domains and proves the theorem. Section 7 places the result in the larger question, with three conjectures that imply the uniqueness conjecture and the computations that support them.

1.2. **Relation to earlier work.** The representations studied here belong to an established connection between Markov numbers, words, and continued fractions. Reutenauer's Christoffel correspondence parametrizes whole Markov triples [18]; the palindromic-closure formula of Reutenauer and Vuillon gives an occurrence-wise construction of their entries [19]. The underlying central-word and incidence identities are developed in [3]. These correspondences distinguish occurrences without presupposing that their largest numerical entries are distinct.

A particularly close arithmetic formulation appears in Çanakçı and Schiffler [6]. They relate primitive sums of two squares to even palindromic continued fractions [6, Theorem 3.10 and Corollary 3.14]. They construct the special continued fraction attached to a Markov occurrence [6, Theorem 4.1], and its right half, through the proof of [6, Corollary 4.4], is the occurrence pair of Proposition 2.2 together with its cost $P+Q-2$, as we show after that proposition. They conjecture the uniqueness of the representation whose palindromification is a Markov snake graph [6, Conjecture 4.5] and identify this conjecture with Markov uniqueness [6, Theorem 4.8]. Their Conjecture 4.6 asks for uniqueness under the weaker requirement that the half continued fraction use only the

digits 1 and 2; they checked both conjectures by computer for all Markov numbers with denominator below 70 [6, Remark 4.7]. Our comparison retains every primitive positive representation of the same integer, with no restriction on its partial quotients. The assertion is that the occurrence-induced representation minimizes the sum of those quotients, with the final Euclidean step omitted. It permits ties and does not assert either of these uniqueness conjectures.

Ordering results compare a different set of objects: the Markov labels at specified lattice points. Aigner's book records conjectures for rows of fixed numerator, denominator and sum [1, Section 10.1]; Rabideau and Schiffler proved the first [17], and Lagisquet, Pelantová, Tavenas, and Vuillon proved all three [14]. Their path minimization holds at a fixed lattice endpoint. Lee, Li, Rabideau, and Schiffler obtain further comparison inequalities [16], which confine any coincidence of labels to a narrow cone of slopes; their fixed-row ratio decreases to a constant, which Section 4.3 computes and bounds on both sides from the Cohn trace, so that each infinite source row lies in a normalized interval. Gaster gives sharp monotonicity ranges using the stable norm on the modular torus [11]. None of these ordering statements, as stated, minimizes Euclidean cost over the full primitive two-square fiber of a label. The classical uniqueness theorem for labels that are prime powers, or twice prime powers, is the case of a fiber with a single element; Remark 3.2 derives it from the results of Sections 2 and 3.

Two of our submitted articles treat neighboring questions. The first studies the same Euclidean cost on the representations produced by a single Gaussian reflection [8], and contains the cost identity of Proposition 2.2 in its source dictionary; the second studies palindromes in the Cohn monoid [9], and contains Lemma 2.5 as the case of equal sums in the proof of its proposition on the lengths of directing words. Their full texts can be consulted in the repository of this series, given at the end of the article.

Related extremal problems fix the length and sum of the entries of a continuant; Kan determines the numerical maximum under these constraints [12, Theorem 5].[1] Our proof derives the required path bounds directly. In its infinite part, a parity order for the full normalized ratio reduces whole boxes of gap lengths to exact corner comparisons. The finitely many remaining labels are settled through their complete Gaussian factorizations, so that an infinite comparison argument is joined to a finite piece of exact arithmetic. The resulting thirteen-row theorem assumes no upper bound on the denominator or on the Markov value. Its injectivity consequence applies when both occurrences belong to the proved rows; it does not exclude a collision with an occurrence outside them.

2. Occurrences, representations, and Euclidean cost

The arithmetic comparison in this paper concerns every primitive representation of a fixed integer, including representations that do not come from a Markov occurrence. We define the cost, put the two sets of objects on the same cost scale, and explain why minimality of the cost is enough to separate two occurrences.

2.1. **The Euclidean cost and its two readings.**

The cost of a pair is read in two ways in this paper: representations are compared through their continued fractions, while the cost of an occurrence is computed from a product of matrices. We define the cost by subtractions and prove that both readings give it. For coprime positive integers r, s, let $\ell(r, s)$ be the number of single subtractions needed to reach $(1, 1)$: subtract the smaller coordinate from the larger until equality. In particular, $\ell(1, 1) = 0$, and exchanging the

[1] For sequences of length t and sum S, the published statement assumes $2 \leqslant t \leqslant S < nt$, where n, a bound on the entries, belongs to the next theorem of that article; the corrected version [13, Theorem 3.5] states the result for $2 \leqslant t \leqslant S$.

coordinates does not change ℓ. For a column v with coprime positive entries we write $\ell(v)$ for the cost of its two entries. Define

$$P_j = \begin{pmatrix} j & 1 \\ 1 & 0 \end{pmatrix} \quad (j \geqslant 1). \tag{1}$$

For every primitive positive column v, exactly j initial subtractions carry $P_j v = (jv_1 + v_2, v_1)^{\mathsf{T}}$ to $(v_2, v_1)^{\mathsf{T}}$. Each of those subtractions is permitted, including the final step when $v_1 = v_2 = 1$. Consequently

$$\ell(P_j v) = j + \ell(v). \tag{2}$$

Lemma 2.1 (The cost by continued fractions and by matrices)**.** *Let $q_0, q_1, \ldots, q_j$ be positive integers, with $q_j \geqslant 2$ unless $j = 0$ and $q_0 = 1$. Then the column*

$$\begin{pmatrix} s \\ r \end{pmatrix} = P_{q_0} P_{q_1} \cdots P_{q_j} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \tag{3}$$

has coprime positive entries, $s/r = [q_0; q_1, \ldots, q_j]$, and

$$\ell(r, s) = q_0 + q_1 + \cdots + q_j - 1. \tag{4}$$

Every pair of coprime integers $0 < r \leqslant s$ arises in this way, from the canonical continued fraction of s/r. If $(r, s) \neq (1, 1)$, then moreover

$$\begin{pmatrix} s \\ r \end{pmatrix} = P_{q_0} \cdots P_{q_{j-1}} P_{q_j - 1} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \tag{5}$$

and $\ell(r, s)$ is the sum of the indices of the matrices in this product.

Proof. The product. We argue by induction on j. If $j = 0$, the column is $P_{q_0}(1, 0)^{\mathsf{T}} = (q_0, 1)^{\mathsf{T}}$, and $s/r = q_0 = [q_0]$. If $j \geqslant 1$, the induction hypothesis applied to $q_1, \ldots, q_j$ gives a column $(s', r')^{\mathsf{T}} = P_{q_1} \cdots P_{q_j}(1, 0)^{\mathsf{T}}$ with positive entries and $s'/r' = [q_1; q_2, \ldots, q_j]$, so that

$$\begin{pmatrix} s \\ r \end{pmatrix} = P_{q_0} \begin{pmatrix} s' \\ r' \end{pmatrix} = \begin{pmatrix} q_0 s' + r' \\ s' \end{pmatrix}, \qquad \frac{s}{r} = q_0 + \frac{r'}{s'} = q_0 + \frac{1}{[q_1; q_2, \ldots, q_j]} = [q_0; q_1, \ldots, q_j].$$

In both cases the entries are positive, and they are coprime because each matrix P_q has determinant -1, so that the column is the image of the primitive column $(1, 0)^{\mathsf{T}}$ under a matrix invertible over $\mathbb{Z}$.

Every pair arises. Let $0 < r \leqslant s$ be coprime. The canonical continued fraction $[q_0; q_1, \ldots, q_j]$ of s/r has positive quotients, since $s/r \geqslant 1$, and its last quotient is at least 2 unless $s/r = 1 = [1]$. The column built from it has coprime positive entries in the ratio s/r, and two primitive positive columns with the same ratio are equal; hence it is $(s, r)^{\mathsf{T}}$.

The cost. For $0 \leqslant i \leqslant j$ put $v_i = P_{q_i} \cdots P_{q_j}(1, 0)^{\mathsf{T}}$, so that $v_0 = (s, r)^{\mathsf{T}}$, $v_j = (q_j, 1)^{\mathsf{T}}$ and $v_i = P_{q_i} v_{i+1}$ for $i < j$. By the first step, every v_i is a primitive positive column, so (2), applied j times, gives

$$\ell(r, s) = q_0 + q_1 + \cdots + q_{j-1} + \ell(q_j, 1).$$

From the pair $(q_j, 1)$ every subtraction lowers the first coordinate by one, and $(1, 1)$ is reached after $q_j - 1$ of them, so that $\ell(q_j, 1) = q_j - 1$. This proves (4).

The second product. If $(r, s) \neq (1, 1)$, then $q_j \geqslant 2$, so that the matrix $P_{q_j - 1}$ is defined, and $P_{q_j - 1}(1, 1)^{\mathsf{T}} = (q_j, 1)^{\mathsf{T}} = P_{q_j}(1, 0)^{\mathsf{T}}$. Substituting this in (3) gives (5). The indices of its factors add up to $q_0 + \cdots + q_{j-1} + (q_j - 1)$, which is $\ell(r, s)$ by (4). □

The introduction described the cost through the continued fraction, as in (4). The proof of Proposition 2.2 below computes it through products of the matrices P_1 and P_2, by (2), as the sum of the indices in (5). Lemma 2.1 shows that the two readings agree, and Lemma 2.3 adds a third, the length of a word.

2.2. **The occurrence-induced pair.** For $0 < P < Q$ with $\gcd(P, Q) = 1$, let $\gamma_{P/Q}$ be the binary word of length Q whose ith letter is b precisely when $\lfloor iP/Q \rfloor - \lfloor (i-1)P/Q \rfloor = 1$. It is the proper lower Christoffel word with

$$|\gamma_{P/Q}|_a = Q - P, \qquad |\gamma_{P/Q}|_b = P. \tag{6}$$

Thus Q is the total length of the Christoffel word. This convention distinguishes the denominator from the number of letters a. The word starts with a and ends with b, so write $\gamma_{P/Q} = azb$. Its interior z is a palindrome. Indeed, for $1 \leqslant j \leqslant Q-1$, coprimality gives $\lfloor (Q-j)P/Q \rfloor = P - 1 - \lfloor jP/Q \rfloor$. The increments defining the letters at positions i and $Q+1-i$ therefore agree for $2 \leqslant i \leqslant Q-1$. This is the central-word property of Christoffel words [3]. We write $\widetilde{w}$ for reversal.

For later exact evaluation, the same occurrence has the Cohn description

$$\mu(a) = \begin{pmatrix} 2 & 1 \\ 1 & 1 \end{pmatrix}, \qquad \mu(b) = \begin{pmatrix} 5 & 2 \\ 2 & 1 \end{pmatrix}, \qquad m_{P/Q} = \frac{1}{3} \operatorname{tr} \mu(\gamma_{P/Q}). \tag{7}$$

The standard Christoffel–Markov correspondence identifies these occurrences with normalized Markov triples having largest entry greater than 2; see, for example, [19, Theorem 1 and its proof]. No injectivity of $P/Q \mapsto m_{P/Q}$ is assumed.

The representation attached to an occurrence, and its cost, are known. Çanakçı and Schiffler show that $m_{P/Q}$ is the numerator of a palindromic continued fraction whose entries are 1 and 2, with $2(Q-P-1)$ entries equal to 1 and $2P$ entries equal to 2, and that $m_{P/Q} = r^2 + s^2$ where s/r is the continued fraction of its right half [6, Section 4.1, Theorem 4.1 and the proof of Corollary 4.4]. That half has partial quotients summing to $(Q-P-1) + 2P = P + Q - 1$ and ending with 2, so Lemma 2.1 gives the cost $P + Q - 2$. The same statement, in the language of Euclidean paths, is the source dictionary of our companion article [8]. Alessandro De Luca and the second author have another proof of the representation, within the combinatorics of words (private communication), in which the Christoffel word of length $m_{P/Q}$ given by the construction of Reutenauer and Vuillon [19] is the image of a word with s letters a and r letters b under a substitution that sends a to a word of length s and b to a word of length r, so that its length is $s^2 + r^2$. We include a proof in the matrix language of the present paper, because it defines the pair that the computations of Section 3 use; the paragraph after the proof shows that it is the pair of Çanakçı and Schiffler.

Proposition 2.2 (Occurrence pair and its cost)**.** *Every occurrence P/Q with $0 < P < Q$ induces a primitive positive representation*

$$m_{P/Q} = r_{P/Q}^2 + s_{P/Q}^2, \qquad 0 < r_{P/Q} < s_{P/Q}.$$

Its subtractive Euclidean cost is $P + Q - 2$.

Proof. The label as a quadratic form. The matrices $\mu(a)$ and $\mu(b)$ are symmetric, so reversing a word transposes its Cohn matrix, and the palindrome z has a symmetric matrix $\mu(z) = \left(\begin{smallmatrix} \alpha & \beta \\ \beta & \delta \end{smallmatrix}\right)$. Multiplying out,

$$\mu(a)\,\mu(z)\,\mu(b) = \begin{pmatrix} 10\alpha + 9\beta + 2\delta & 4\alpha + 4\beta + \delta \\ 5\alpha + 7\beta + 2\delta & 2\alpha + 3\beta + \delta \end{pmatrix},$$

whose trace is $3(4\alpha+4\beta+\delta)$. With $e=(2,1)^{\mathsf T}$ one has $e^{\mathsf T}\mu(z)e=4\alpha+4\beta+\delta$, and therefore

$$m_{P/Q}=\tfrac{1}{3}\operatorname{tr}\mu(\gamma_{P/Q})=e^{\mathsf T}\mu(z)\,e. \tag{8}$$

A square root of the form. Cut the palindrome at its middle: $z=\widetilde{w}\,w$ when $|z|$ is even, and $z=\widetilde{w}\,a\,w$ or $z=\widetilde{w}\,b\,w$ when $|z|$ is odd, according to its middle letter. Since $\mu(\widetilde{w})=\mu(w)^{\mathsf T}$, $\mu(a)=P_1^2$ and $\mu(b)=P_2^2$, with P_1 and P_2 symmetric, we obtain $\mu(z)=H^{\mathsf T}H$, where H is $\mu(w)$, $P_1\mu(w)$ or $P_2\mu(w)$ in the three cases; the empty palindrome gives $H=I$. By (8), $m_{P/Q}=(He)^{\mathsf T}(He)$.

The pair. The integer matrix H has determinant ±1, so the column He is primitive. Its entries are positive, because H is a product of nonnegative matrices with no zero row and e is positive. They are distinct: two equal coprime entries would both be 1, giving $m_{P/Q}=2$, whereas $m_{P/Q}\geqslant 5$, because $\mu(z)\geqslant I$ entrywise (a product of nonnegative matrices each at least I) and (8) then gives $m_{P/Q}\geqslant e^{\mathsf T}e=5$. Sorting the two entries defines $r_{P/Q}<s_{P/Q}$. The construction uses the occurrence alone and assumes nothing about other occurrences with the same label.

Its cost. The column e reaches $(1,1)$ after one subtraction, so $\ell(e)=1$. Apply the factors of H to e from the right. By (2), a factor $\mu(a)=P_1P_1$ coming from a letter a of w adds 2 to the cost, a factor $\mu(b)=P_2P_2$ adds 4, and the middle factor P_1 or P_2, when present, adds 1 or 2. Each letter of w occurs twice in z, so these increments add up to $|z|_a+2|z|_b$, and (6) gives

$$\ell(He)=1+|z|_a+2|z|_b=1+(Q-P-1)+2(P-1)=P+Q-2.$$

No restriction on the parity of $|z|$ or of $m_{P/Q}$ is needed. □

The pair of Çanakçı and Schiffler is this pair. Write $\gamma_{P/Q}=a^{u_1}b\,a^{u_2}b\cdots a^{u_P}b$. The kth letter b stands at position $\lceil kQ/P\rceil$, and kQ/P is not an integer for $1\leqslant k<P$; comparison with their crossing numbers v_k, which satisfy $v_1+\cdots+v_k=\lfloor kQ/P\rfloor$ for $k<P$ and $v_1+\cdots+v_P=Q-1$, gives $u_1=v_1$ and $u_k=v_k-1$ for $2\leqslant k\leqslant P$. Their palindromic continued fraction is therefore $[2,\sigma(z),2]$, where σ replaces each letter a by the two entries $1,1$ and each letter b by $2,2$. Its right half is $[\sigma(w),2]$, or $[j,\sigma(w),2]$ with $j=1$ or 2 for a middle letter a or b, and since $\mu(a)=P_1^2$, $\mu(b)=P_2^2$ and $e=P_2(1,0)^{\mathsf T}$, this is the continued fraction of the column He.

2.3. **Every primitive pair has a unique Euclidean directive.** Introduce the elementary matrices

$$L_a=\begin{pmatrix}1&1\\0&1\end{pmatrix},\qquad L_b=\begin{pmatrix}1&0\\1&1\end{pmatrix},\qquad L_{uv}=L_uL_v,\qquad \mathbf{1}=\begin{pmatrix}1\\1\end{pmatrix}. \tag{9}$$

Lemma 2.3 (Euclidean bijection)**.** *The map $u\mapsto L_u\mathbf{1}$ is a bijection from binary words to ordered coprime positive pairs, and*

$$\ell(L_u\mathbf{1})=|u|. \tag{10}$$

Proof. If $x>y>0$, the first letter of any directive producing (x,y) must be a, because L_a maps positive pairs to pairs with first coordinate larger, whereas L_b makes the second coordinate larger. The unique predecessor is $(x-y,y)$. For $y>x$, the first letter is b and the predecessor is $(x,y-x)$. Each subtraction preserves positivity and coprimality and decreases $x+y$. The only equal coprime positive pair is $(1,1)$, so this process terminates there. Reading the uniquely determined letters gives both existence and uniqueness, and one letter is removed at each subtraction. This proves (10). □

The word u is the *directive* of the pair $L_u\mathbf{1}$, and it can be read on the continued fraction. With $J=\left(\begin{smallmatrix}0&1\\1&0\end{smallmatrix}\right)$, which exchanges the two coordinates, one has $P_q=L_a^qJ$,

$JL_a^q J = L_b^q$ and $J\mathbf{1} = \mathbf{1}$. Grouping the factors of (5) in consecutive pairs therefore gives, for coprime $0 < r < s$ with $s/r = [q_0; q_1, \dots, q_j]$,

$$\begin{pmatrix} s \\ r \end{pmatrix} = L_a^{q_0} L_b^{q_1} L_a^{q_2} L_b^{q_3} \cdots \mathbf{1},$$

where the exponents are $q_0, q_1, \dots, q_{j-1}, q_j - 1$ and the letters alternate. The directive of (s, r) is thus the word $a^{q_0} b^{q_1} a^{q_2} \cdots$ whose runs have these lengths, and its length $q_0 + \cdots + q_j - 1$ agrees with (4) and (10). For instance $21/13 = [1; 1, 1, 1, 1, 2]$, and the directive of $(21, 13)$ is *ababab*.

For unordered representations we always choose $0 < r < s$. This loses neither norms nor costs. The exceptional positive pair $(1, 1)$ represents 2 and can be treated separately; $1 = 1^2 + 0^2$ is outside the positive pair convention.

Definition 2.4. An occurrence P/Q is *weakly Euclidean-minimal* if

$$P + Q - 2 \leqslant \ell(r, s) \quad \text{whenever} \quad m_{P/Q} = r^2 + s^2, \quad 0 < r < s, \quad \gcd(r, s) = 1. \tag{11}$$

The competitors in (11) range over the entire primitive two-square fiber. Their Euclidean directives need not be directives induced by a central Cohn word as above. Ties with such other representations are allowed.

2.4. **Why weak minimality is enough within a proved family.** Among the ordering theorems recalled in the introduction, we use the fixed-sum theorem of Lagisquet, Pelantová, Tavenas and Vuillon [14, Theorem 2, item 3], which compares two lattice points differing by a unit step along a line of fixed sum. Applying it i times gives, for reduced P/Q and $(P - i)/(Q + i)$ with $0 < i < P < Q$,

$$m_{P/Q} < m_{(P-i)/(Q+i)}. \tag{12}$$

The intermediate points of that iteration need not be reduced; this is harmless because the authors define the label at every lattice point, not only at Farey fractions. The inequality compares distinct points on one line of fixed sum $P + Q$. It does not compare arbitrary two-square representations.

The next lemma is an immediate consequence of (12); it is also the case of equal sums in the proof of the proposition on the lengths of directing words of Markov words in [9], where the cost $P + Q - 2$ appears as the length of a directing word.

Lemma 2.5 (Fixed-cost injectivity)**.** *Two occurrences having the same label and the same occurrence cost have the same reduced Farey coordinate.*

Proof. Their costs are $P + Q - 2$ and $P' + Q' - 2$. If they are equal and $P > P'$, set $i = P - P'$. Then $Q' = Q + i$ and $0 < i < P$, so (12) contradicts equality of labels. The other strict ordering of the numerators is excluded in the same way. Consequently $P = P'$ and $Q = Q'$. A reduced Farey coordinate has a unique lower Christoffel word, hence a unique normalized occurrence. □

Corollary 2.6. *The label map is injective on any family of occurrences all of which are weakly Euclidean-minimal.*

Proof. If two members have equal label, each occurrence-induced pair is a competitor in the other's two-square fiber. Weak minimality in both directions makes their costs equal. Apply Lemma 2.5. □

Thus weak minimality on every occurrence would imply the Frobenius–Markov uniqueness conjecture. For a restricted family, both occurrences must lie in that family. If only one is known to be weakly minimal, a distinct equal-label occurrence is merely forced to have strictly larger cost.

3. Complete Gaussian fibers and the finite windows

The finite part of the proof must compare an occurrence with every primitive representation of its label. Searching a coordinate rectangle would obscure completeness. Gaussian factorization gives an exact finite parameter space whose size depends on the distinct prime divisors, not on the size of the coordinates, and once the prime factorizations are known the enumeration is a matter of integer arithmetic alone.

3.1. The whole primitive representation fiber.

For $M \geqslant 5$, define

$$\mathcal{F}(M) = \{(x, y) \in \mathbb{Z}^2 : 0 < x < y,\ \gcd(x, y) = 1,\ x^2 + y^2 = M\}.$$

The source construction in Section 2 proves that this set is nonempty for every occurrence considered here.

Proposition 3.1 (Primitive Gaussian allocation)**.** *If $\mathcal{F}(M) \neq \varnothing$, then*

$$M = 2^{\epsilon} \prod_{j=1}^{t} p_j^{e_j}, \qquad \epsilon \in \{0, 1\}, \quad t \geqslant 1, \quad p_j \equiv 1 \pmod 4, \tag{13}$$

where the p_j are distinct primes and $e_j \geqslant 1$. Choose $\pi_j = a_j + ib_j$ with $p_j = a_j^2 + b_j^2$ and $0 < a_j < b_j$. The elements of $\mathcal{F}(M)$ are exactly the sorted absolute coordinate pairs of

$$(1 + i)^{\epsilon} \prod_{j=1}^{t} \begin{cases} \pi_j^{e_j}, & \sigma_j = 1, \\ \overline{\pi_j}^{e_j}, & \sigma_j = -1, \end{cases} \qquad (\sigma_1, \ldots, \sigma_t) \in \{1, -1\}^t. \tag{14}$$

Each normalized pair occurs exactly twice. Equivalently, fixing $\sigma_1 = 1$ lists the fiber without repetition, and $|\mathcal{F}(M)| = 2^{t-1}$.

Proof. The Gaussian integers form a Euclidean domain for the squared norm, hence have unique factorization. If $z = x + iy$ is primitive and a rational prime $p \equiv 3 \pmod 4$ divides $z\overline{z}$, its Gaussian primality forces it to divide z or $\overline{z}$. Either condition gives $p \mid x, y$, a contradiction. Likewise $4 \nmid M$: a sum of two squares divisible by four has both coordinates even. This proves the shape (13); $t = 0$ would leave $M = 1$ or 2.

For a split prime $p_j = \pi_j\overline{\pi_j}$, an allocation using both π_j and $\overline{\pi_j}$ in z would make p_j divide both coordinates. Primitivity therefore puts the whole exponent e_j on one of the two conjugate factors. Conversely, each allocation in (14) has primitive coordinates: a common odd prime would require both conjugate Gaussian factors, and a common factor two would require a higher ramified exponent. Unique factorization proves that these allocations, up to multiplication by a unit, exhaust all representations. Since $M \geqslant 5$, a primitive Gaussian integer of norm M has nonzero coordinates of distinct absolute values, for otherwise $M = 1$ or $M = 2$; its sorted absolute pair therefore lies in $\mathcal{F}(M)$.

It remains to count. Two Gaussian integers $x+iy$ and $x'+iy'$ have the same normalized pair, that is $\{|x|, |y|\} = \{|x'|, |y'|\}$, exactly when $x' + iy'$ is $u(x + iy)$ or $u(x - iy)$ for one of the four units u. Write z_σ for the product (14). For an odd prime $p_j = a_j^2 + b_j^2$ the factors π_j and $\overline{\pi_j}$ are not associate: $\overline{\pi_j} = u\pi_j$ would force $b_j = 0$, $a_j = 0$ or $a_j = \pm b_j$, none of which is possible since $0 < a_j < b_j$. Suppose now that $z_{\sigma'}$ and z_σ give the same normalized pair. If $z_{\sigma'} = uz_\sigma$, unique factorization compares the exponents of the non-associate primes π_j and $\overline{\pi_j}$ and gives $\sigma' = \sigma$. If $z_{\sigma'} = u\overline{z_\sigma}$, the same comparison gives $\sigma' = -\sigma$, because conjugation exchanges π_j and $\overline{\pi_j}$ and replaces $1 + i$ by the associate $1 - i = -i(1 + i)$. Conversely $z_{-\sigma}$ is an associate of $\overline{z_\sigma}$, so σ and $-\sigma$ always give the same pair. As $t \geqslant 1$, the vectors σ and $-\sigma$ are distinct, so each normalized pair comes from exactly two sign vectors, which proves the count and the last assertion. □

The exponents in (13) do not enlarge the primitive sign cube. Mixed allocations of a prime power generate imprimitive representations, so omitting them is a proved reduction, not a filter based on observed costs.

Remark 3.2 (Labels with a single representation)**.** By Proposition 3.1, a label $m > 2$ has exactly one primitive positive representation precisely when $t = 1$, that is when $m = p^e$ or $m = 2p^e$ for a prime $p \equiv 1 \pmod 4$. Such a label belongs to exactly one reduced fraction. *Step 1.* Let $m = m_{P/Q} = m_{P'/Q'}$; by Proposition 2.2 the two occurrence pairs are elements of $\mathcal{F}(m)$, of costs $P+Q-2$ and $P'+Q'-2$. *Step 2.* Since $|\mathcal{F}(m)| = 2^{t-1} = 1$, the two pairs coincide, and so do their costs: $P + Q = P' + Q'$. *Step 3.* Lemma 2.5 gives $P/Q = P'/Q'$. This recovers the theorem of Baragar, Button and Schmutz that a Markov number which is a prime power or twice a prime power is unique [2, 5, 20], in the form stated in [22, Theorem C]; see also [15]. Here it follows from the fixed-sum ordering (12) and the Gaussian count, and Theorem 1.1 extends the same fiber argument, on thirteen rows, to fibers of every size.

Example 3.3 (Two representations, two costs)**.** The first-row label $m_{1/7} = 610 = 2\cdot 5\cdot 61$ has exactly two primitive positive representations:

$$610 = 13^2 + 21^2 = 9^2 + 23^2.$$

In the normalization of the proposition the two split primes are $1+2i$ and $5+6i$, and the two allocations give $(1+i)(1+2i)(5+6i) = -23+9i$ and $(1+i)(1-2i)(5+6i) = 21+13i$. Proposition 3.1 proves that the list is complete. Their continued fractions are

$$\frac{21}{13} = [1; 1, 1, 1, 1, 2], \qquad \frac{23}{9} = [2; 1, 1, 4],$$

so the Euclidean costs are 6 and 7, respectively. The occurrence cost $1 + 7 - 2 = 6$ is therefore the minimum. This illustrates why the optimization question persists even when the label and its Markov occurrence are fixed: the full arithmetic fiber contains another primitive pair whose directive is unrestricted.

Proposition 3.1 lists the whole fiber but does not say which of its elements comes from the occurrence. The two labels next to the occurrence in the Markov tree do: the occurrence pair is the element of the sign cube whose prime factors all divide one explicit Gaussian integer.

Proposition 3.4 (The occurrence pair in the sign cube)**.** *Let $0 < P < Q$ be coprime, let $L = P_L/Q_L < R = P_R/Q_R$ be its two Farey parents, the reduced fractions with $P_L + P_R = P$, $Q_L + Q_R = Q$ and $P_RQ_L - P_LQ_R = 1$, and let m_L, m_R be their labels, with $m_{0/1} = 1$ and $m_{1/1} = 2$. Write $(s, r)^{\mathsf{T}} = He$ for the occurrence pair of P/Q before sorting, and $m = m_{P/Q}$. Then the ideal of $\mathbb{Z}[i]$ generated by m and $m_L + i\,m_R$ is generated by $s + ir$ or by its conjugate. In particular, in the sign cube (14) of m, the occurrence pair is the element whose prime factors all divide $m_L + i\,m_R$, up to the conjugation of the whole product.*

Proof. The root read from the Cohn matrix. The lower Christoffel word of P/Q is the product of those of its two Farey parents, the words of $0/1$ and $1/1$ being the letters a and b [18]. Writing γ_L and γ_R for the words of the parents, we have $\mu(\gamma_{P/Q}) = \mu(\gamma_L)\,\mu(\gamma_R)$. Each factor has determinant one and upper right entry equal to its label: for the letters, $\mu(a)$ and $\mu(b)$ show it, and for a longer Christoffel word the computation in the proof of Proposition 2.2 gives an upper right entry $4\alpha + 4\beta + \delta$, which is the label by (8). Writing $\mu(\gamma_{P/Q}) = \left(\begin{smallmatrix} a & m \\ c & d \end{smallmatrix}\right)$ and $\mu(\gamma_X) = \left(\begin{smallmatrix} a_X & m_X \\ c_X & d_X \end{smallmatrix}\right)$ for $X = L, R$, the upper row of the product gives $m = a_L m_R + m_L d_R$ and $a = a_L a_R + m_L c_R$, so that

$$a\,m_R = a_R(m - m_L d_R) + m_L c_R m_R \equiv -m_L(a_R d_R - m_R c_R) = -m_L \pmod m.$$

The root read from the occurrence pair. By the proof of Proposition 2.2, $\mu(\gamma_{P/Q}) = \mu(a)H^{\mathsf{T}}H\mu(b)$. With $e_1 = (1,0)^{\mathsf{T}}$, the symmetric matrix $\mu(a)$ satisfies $\mu(a)e_1 = e$, and $\mu(b)e_1 = (5,2)^{\mathsf{T}} = 2e + e_1$. Put $v = He = (s,r)^{\mathsf{T}}$ and $h = He_1 = (p,q)^{\mathsf{T}}$. Then

$$a = e_1^{\mathsf{T}}\mu(\gamma_{P/Q})e_1 = (He)^{\mathsf{T}}H(2e + e_1) = 2m + sp + rq.$$

The matrix with columns v and h is HP_2, of determinant $\eta = sq - rp = \pm 1$, so that $rp = sq - \eta$ and

$$a\,r \equiv (sp + rq)\,r = s(sq - \eta) + r^2 q = qm - \eta s \equiv -\eta s \pmod{m}.$$

The ideals. The labels m_L, m_R, m form a Markov triple, whose entries are pairwise coprime [1]; and $\gcd(s,r) = 1$ with $s^2 + r^2 = m$. Hence m_R and r are invertible modulo m. By the first step, $m_L + i\,m_R \equiv m_R\,(i - a) \pmod{m}$, and by the second, $s \equiv -\eta a r$, so that $s + ir \equiv r\,(i - \eta a)$ and $s - ir \equiv -r\,(i + \eta a) \pmod{m}$. If $\eta = 1$, the first congruence and the third give equal ideals $(m, s + ir) = (m, i - a) = (m, m_L + i\,m_R)$; if $\eta = -1$, the fourth does, $(m, s - ir) = (m, m_L + i\,m_R)$. Finally $(m, s \pm ir) = (s \pm ir)$, because $s \pm ir$ divides $m = (s + ir)(s - ir)$.

The sign cube. The generator $s \pm ir$ of the ideal is the greatest common divisor of m and $m_L + i\,m_R$. A Gaussian prime factor of the label divides it exactly when it divides $m_L + i\,m_R$, which identifies the allocation of (14), up to the conjugation of the whole product. □

The congruence $a\,m_R \equiv -m_L \pmod{m}$ of the first step is classical: up to sign, the residue of a modulo m is the characteristic number of the Markov triple (m, m_L, m_R), defined by Frobenius as the solution of $m_R x \equiv \pm m_L \pmod{m}$, a square root of -1 modulo m [10, §2, p. 462], see also [1, Section 2.2]. The proposition translates it into the Gaussian integers and identifies the corresponding element of the sign cube with the occurrence pair. The cofactor $(m_L + i\,m_R)/(s \pm ir)$ has norm $(m_L^2 + m_R^2)/m = 3m_L m_R - m$, by the Markov equation, in the form $mm' = m_L^2 + m_R^2$ with $m' = 3m_L m_R - m$ that Frobenius uses [10, §1, p. 460]. For the label 610 of the example above, the parents 0/1 and 1/6 have labels 1 and 233, and

$$1 + 233\,i = (21 + 13\,i)(5 + 8\,i)\,:$$

the first factor is the occurrence pair $(1 + i)(1 - 2i)(5 + 6i)$, and the second has norm $89 = 3 \cdot 1 \cdot 233 - 610$, the label of 1/5. The second factor is no accident: 1/5 is the third vertex of the Farey triangle $(0/1, 1/6, 1/5)$ of 1/7, defined below, and the next subsection shows that the cofactor is always the occurrence pair of that vertex, up to a unit and a conjugation; here $5 + 8i = i\,(8 - 5i)$, and $(8, 5)$ is the occurrence pair of 1/5.

3.2. **The occurrence pairs along the Farey tree.** Proposition 3.4 finds the occurrence pair as a greatest common divisor, a computation that still involves the label itself. The Markov tree offers a recursion instead. Along the tree the labels obey two relations of Vieta, a sum and a product, and the product lifts exactly to the Gaussian integers. The Gaussian integer $m_L + i\,m_R$ of Proposition 3.4 is the product of two Gaussian integers attached to P/Q and to the third vertex of its Farey triangle, which are their occurrence pairs up to units and conjugation; once an orientation is fixed at the root of the tree, the product holds exactly, without any unit. Every occurrence pair is thus obtained from pairs found earlier by one exact division in $\mathbb{Z}[i]$, with no factorization, no greatest common divisor and no continued fraction. This is the Gaussian form of the replacement $(x, y, z) \mapsto (x, y, 3xy - z)$ that generates the Markov tree.

Two fractions $L = P_L/Q_L < R = P_R/Q_R$ with nonnegative terms are *Farey neighbors* when $P_R Q_L - P_L Q_R = 1$. Their mediant is $M = (P_L + P_R)/(Q_L + Q_R)$, and we call their *difference* the fraction D whose terms are $|P_R - P_L|$ and $|Q_R - Q_L|$. The Farey

tree of $[0,1]$ starts from the pair $(0/1, 1/1)$, whose difference is $1/0$, and replaces each pair (L,R) by the two pairs (L,M) and (M,R). Every reduced fraction of $(0,1)$ is the mediant of exactly one pair of the tree, the pair of its two Farey parents [1, Section 3.2, Theorem 3.9 and its proof]. The differences of the two new pairs are R and L, because $M-L=R$ and $R-M=-L$ term by term, so that the difference of a pair of the tree is either one of the three fractions $0/1$, $1/1$ and $1/0$, or a mediant formed earlier. We write γ_X and m_X for the lower Christoffel word and the label of a fraction X of $[0,1]$, with $\gamma_{0/1}=a$ and $\gamma_{1/1}=b$, we set $m_{1/0}=1$, and we call (L,R,D) the *Farey triangle* of the mediant M. Finally, $N(g+ih)=g^2+h^2$ denotes the norm of $\mathbb{Z}[i]$.

The first lemma records the two relations of Vieta on every triangle. In this indexing they are the content of Theorem 1 of Reutenauer and of its proof [18, Theorem 1 and its proof], where the Fricke identity gives the Markov triple of a standard factorization and two trace identities give the two cases below; we include the argument in the notation of the Farey tree.

Lemma 3.5 (The labels on a Farey triangle)**.** *For every pair (L,R) of the tree, with mediant M and difference D, the triple (m_L, m_R, m_D) is a Markov triple with pairwise coprime entries, and*

$$m_M = 3m_Lm_R - m_D, \qquad m_Mm_D = m_L^2 + m_R^2. \tag{15}$$

Proof. The sum relation. Put $A_X=\mu(\gamma_X)$ for every fraction $X\neq 1/0$ of the tree. Since $\gamma_M=\gamma_L\gamma_R$ [18], we have $A_M=A_LA_R$, and each A_X has determinant one and trace $3m_X$ by (7). For two matrices A and B of determinant one, the Cayley–Hamilton theorem gives $B+B^{-1}=(\operatorname{tr}B)\,I$; multiplying by A and taking traces,

$$\operatorname{tr}(AB)+\operatorname{tr}(AB^{-1})=\operatorname{tr}A\,\operatorname{tr}B.$$

With $A=A_L$ and $B=A_R$ this reads $3m_M+\operatorname{tr}(A_LA_R^{-1})=9m_Lm_R$, and it remains to show that $\operatorname{tr}(A_LA_R^{-1})=3m_D$. For the starting pair,

$$A_{0/1}A_{1/1}^{-1}=\begin{pmatrix}2&1\\1&1\end{pmatrix}\begin{pmatrix}1&-2\\-2&5\end{pmatrix}=\begin{pmatrix}0&1\\-1&3\end{pmatrix},$$

whose trace is $3=3m_{1/0}$. Every other pair (L,R) was created from a pair of the tree in one of two ways. Either it is (L,M'), where $M'=R$ is the mediant of a pair (L,R'); then $D=R'$ and $A_R=A_LA_D$, so that $A_LA_R^{-1}=A_LA_D^{-1}A_L^{-1}$, whose trace is $\operatorname{tr}A_D^{-1}=\operatorname{tr}A_D=3m_D$, a matrix of determinant one and its inverse having the same trace. Or it is (M',R), where $M'=L$ is the mediant of a pair (L',R); then $D=L'$ and $A_L=A_DA_R$, so that $A_LA_R^{-1}=A_D$, whose trace is $3m_D$.

The Markov triples. The starting triple $(1,2,1)$ satisfies $x^2+y^2+z^2=3xyz$. If (m_L,m_R,m_D) satisfies it, then m_D is a root of the polynomial $X^2-3m_Lm_RX+m_L^2+m_R^2$, and by the sum relation $m_M=3m_Lm_R-m_D$ is its other root. The product of the two roots is $m_Mm_D=m_L^2+m_R^2$, which is the second relation in (15), and (m_L,m_R,m_M) satisfies the Markov equation as well. The triples of the two new pairs, (m_L,m_M,m_R) and (m_M,m_R,m_L), are permutations of it, and induction on the depth in the tree proves that every triple (m_L,m_R,m_D) is a Markov triple.

Coprimality. If a prime divides two entries of a Markov triple, the Markov equation shows that it divides the square of the third, hence the third. The triple of a new pair has the entries m_L, m_R and m_M, so a prime dividing two of them would divide m_L and m_R, two entries of the triple of the parent pair. Since the starting triple $(1,2,1)$ has pairwise coprime entries, induction on the depth proves the last assertion. □

The product relation is the one to lift. Put $\alpha_{0/1} = 1$, $\alpha_{1/1} = 1 + i$ and $\alpha_{1/0} = 1$, and, for each pair (L, R) of the tree, with mediant M and difference D,

$$\overline{\alpha_M} = \frac{m_L + i\, m_R}{\alpha_D}. \tag{16}$$

The definition proceeds down the tree: when a pair is reached, α_D is already known, since D is a starting fraction or a mediant formed earlier, and it remains to see that the division is exact. Three divisibilities attached to each triangle ensure it. They are exchanged by the circular permutation $L \to R \to D \to L$: the first says that the division (16) is exact, and the other two prepare the divisions of the next generation. The first lemma below carries them from a pair to its two new pairs; the second runs the induction down the tree.

Lemma 3.6 (One step down the tree)**.** *Let (L, R) be a pair of the tree, with mediant M and difference D, and let α_L, α_R, α_D be Gaussian integers of norms m_L, m_R, m_D such that*

$$\alpha_D \mid m_L + i\, m_R, \qquad \alpha_L \mid m_R + i\, m_D, \qquad \alpha_R \mid m_D + i\, m_L. \tag{17}$$

Then the quotient (16) is a Gaussian integer, $N(\alpha_M) = m_M$, and

$$\alpha_M \mid m_R + i\, m_L, \qquad \alpha_L \mid m_M + i\, m_R, \qquad \alpha_R \mid m_L + i\, m_M. \tag{18}$$

Proof. The norm of α_M. The first divisibility in (17) makes the quotient (16) a Gaussian integer, and by Lemma 3.5,

$$N(\alpha_M) = \frac{N(m_L + i\, m_R)}{N(\alpha_D)} = \frac{m_L^2 + m_R^2}{m_D} = \frac{m_M m_D}{m_D} = m_M.$$

Divisibility by α_M. Conjugating (16) gives $\alpha_M \overline{\alpha_D} = m_L - i\, m_R$, and multiplying by i,

$$\alpha_M \cdot i\, \overline{\alpha_D} = m_R + i\, m_L.$$

Divisibility by α_L. We compute modulo α_L, which divides $m_L = \alpha_L \overline{\alpha_L}$. The definition gives $\overline{\alpha_M} \alpha_D = m_L + i\, m_R \equiv i\, m_R$, and its conjugate gives $\alpha_M \overline{\alpha_D} = m_L - i\, m_R$. With $m_M = \alpha_M \overline{\alpha_M}$ and $m_D = \alpha_D \overline{\alpha_D}$ it follows that, modulo α_L,

$$\begin{aligned} m_M + i\, m_R &\equiv \alpha_M \overline{\alpha_M} + \overline{\alpha_M} \alpha_D = \overline{\alpha_M}\,(\alpha_M + \alpha_D), \\ \overline{\alpha_D}\,(\alpha_M + \alpha_D) &= m_L - i\, m_R + m_D \equiv m_D - i\, m_R = -i\,(m_R + i\, m_D). \end{aligned}$$

The last member is divisible by α_L, by the second divisibility in (17). The Gaussian integers α_L and $\overline{\alpha_D}$ have no common prime factor, because their norms m_L and m_D are coprime by Lemma 3.5. Unique factorization in $\mathbb{Z}[i]$ therefore gives $\alpha_L \mid \alpha_M + \alpha_D$, and then $\alpha_L \mid m_M + i\, m_R$ by the first line.

Divisibility by α_R. We compute modulo α_R, which divides m_R. The definition gives $\overline{\alpha_M} \alpha_D = m_L + i\, m_R \equiv m_L$, and therefore, modulo α_R,

$$\begin{aligned} m_L + i\, m_M &\equiv \overline{\alpha_M} \alpha_D + i\, \alpha_M \overline{\alpha_M} = \overline{\alpha_M}\,(\alpha_D + i\, \alpha_M), \\ \overline{\alpha_D}\,(\alpha_D + i\, \alpha_M) &= m_D + i\,(m_L - i\, m_R) = m_D + m_R + i\, m_L \equiv m_D + i\, m_L. \end{aligned}$$

The last member is divisible by α_R, by the third divisibility in (17). The norms m_R and m_D are coprime, so α_R and $\overline{\alpha_D}$ have no common prime factor, and unique factorization gives $\alpha_R \mid \alpha_D + i\, \alpha_M$, and then $\alpha_R \mid m_L + i\, m_M$ by the first line. □

Lemma 3.7 (Three divisibilities)**.** *Every quotient (16) is a Gaussian integer, and every pair (L, R) of the tree, with difference D, satisfies (17) and*

$$N(\alpha_L) = m_L, \qquad N(\alpha_R) = m_R, \qquad N(\alpha_D) = m_D. \tag{19}$$

Proof. We argue by induction on the depth of the pair in the tree.

The starting pair. For $(L, R, D) = (0/1, 1/1, 1/0)$, the integers 1, $1+i$ and 1 have the norms 1, 2 and 1. The first two divisibilities have the divisor 1, and the third says that $1+i$ divides $m_{1/0} + i\, m_{0/1} = 1+i$.

The step. Assume (19) and (17) for a pair (L, R). By Lemma 3.6, the quotient (16) is a Gaussian integer of norm m_M, and (18) holds. The pair (L, M) has difference R, and its three divisibilities read $\alpha_R \mid m_L + i\, m_M$, $\alpha_L \mid m_M + i\, m_R$ and $\alpha_M \mid m_R + i\, m_L$; the pair (M, R) has difference L, and its three divisibilities read $\alpha_L \mid m_M + i\, m_R$, $\alpha_M \mid m_R + i\, m_L$ and $\alpha_R \mid m_L + i\, m_M$. Both are the three divisibilities (18), and the norms of the two new pairs are among m_L, m_R and $m_M = N(\alpha_M)$. Both new pairs therefore satisfy (19) and (17), which completes the induction. □

The lemma defines α_M for every reduced fraction M of $(0, 1)$, since M is the mediant of exactly one pair of the tree. It remains to identify this Gaussian integer.

Proposition 3.8 (The Vieta jump in the Gaussian integers)**.** *Let $M = P/Q$ be a reduced fraction of $(0, 1)$ with Farey triangle (L, R, D). Then*

$$m_L + i\, m_R = \overline{\alpha_M}\, \alpha_D, \qquad N(\alpha_M) = m_M, \tag{20}$$

and α_M is the occurrence pair of P/Q up to a unit and a conjugation: if $(s, r)^{\mathsf{T}} = He$, then $\alpha_M = \varepsilon(s + ir)$ or $\alpha_M = \varepsilon(s - ir)$ for a unit ε of $\mathbb{Z}[i]$. In particular, when D lies in $(0, 1)$, the cofactor α_D of $\overline{\alpha_M}$ in $m_L + i\, m_R$ is the occurrence pair of D, up to a unit and a conjugation.

Proof. Lemma 3.7 shows that (16) is a Gaussian integer of norm m_M, and multiplying (16) by α_D gives the product in (20). The Gaussian integer $\overline{\alpha_M}$ divides $m_M = \overline{\alpha_M}\alpha_M$ and $m_L + i\, m_R = \overline{\alpha_M}\alpha_D$, hence every element of the ideal of $\mathbb{Z}[i]$ generated by m_M and $m_L + i\, m_R$. By Proposition 3.4, this ideal is generated by $s + ir$ or by $s - ir$, so that $\overline{\alpha_M}$ divides one of them. The two sides of this divisibility have the same norm m_M, so their quotient is a Gaussian integer of norm one, a unit: $\overline{\alpha_M} = \varepsilon'(s \pm ir)$, and conjugation gives $\alpha_M = \overline{\varepsilon'}(s \mp ir)$. The last assertion is the first one applied to the fraction D, which has its own Farey triangle. □

Up to units and conjugation, the product form is classical in substance: it follows from the identity $mm' = m_L^2 + m_R^2$ and from the characteristic numbers of Frobenius, both recalled after Proposition 3.4 and applied to m and to $m' = m_D$, together with unique factorization in $\mathbb{Z}[i]$ [10, §§1–2]. For m_D one more step is needed, since (m_D, m_L, m_R) is not the triple of the Farey parents of D: the Farey triangles with the vertex D are joined by flips that keep D, each of which replaces a triple (m_D, x, y) by $(m_D, y, 3m_D y - x)$, and this leaves the residue of x/y modulo m_D unchanged up to sign, because $3m_D y - x \equiv -x$ and $(x/y)^2 \equiv -1 \pmod{m_D}$ by the Markov equation. What Lemmas 3.6 and 3.7 add is the exact orientation, without units, along the whole tree, and it is this orientation that makes the recursion possible.

The proposition turns the computation of occurrence pairs into a descent. To obtain the occurrence pair of P/Q, one follows the path of P/Q in the Farey tree and keeps only the three Gaussian integers of the current triangle; each step costs one exact division in $\mathbb{Z}[i]$. Figure 2 shows the first four generations. For the triangle $L = 1/2$, $R = 2/3$, $D = 1/1$, whose labels are 5, 29 and 2, the rule (16) gives

$$\overline{\alpha_{3/5}} = \frac{5 + 29\, i}{1 + i} = \frac{(5 + 29\, i)(1 - i)}{2} = \frac{34 + 24\, i}{2} = 17 + 12\, i,$$

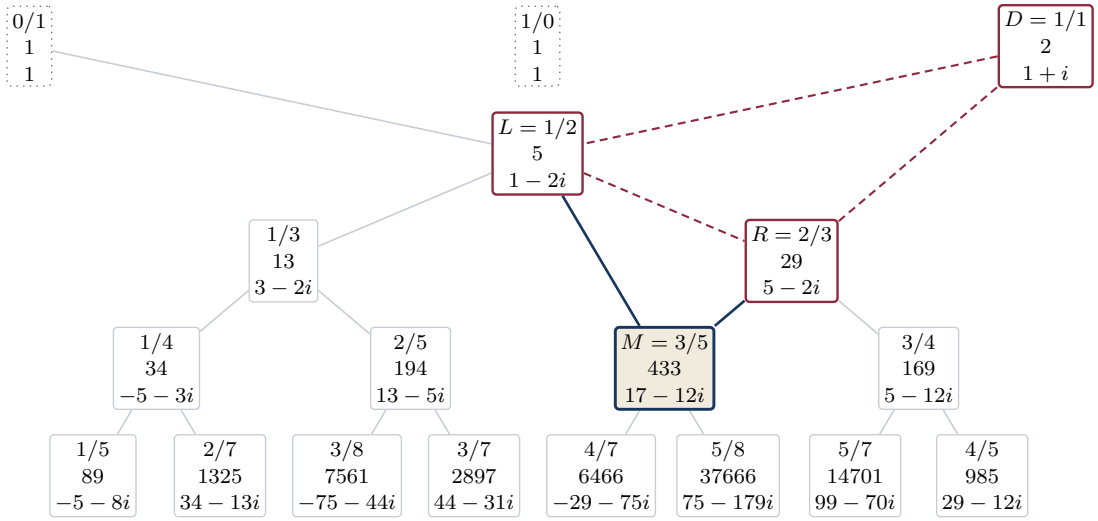


FIGURE 2. The first four generations of the Farey tree, with the label m and the Gaussian integer α of each fraction, computed by the rule (16) alone from the values at the three starting fractions $0/1$, $1/1$ and $1/0$, drawn at the top. Each fraction is joined to the younger of its two Farey parents, and $1/2$ to both. In wine and dashed, the Farey triangle $L = 1/2$, $R = 2/3$, $D = 1/1$; in navy, the edges from its parents to the mediant $M = 3/5$, whose Gaussian integer $17 - 12i$ is the conjugate of the quotient of $5 + 29i$ by $1 + i$.

and indeed $17^2 + 12^2 = 433 = m_{3/5}$. Along the fractions $P/(P+1)$, whose right parent is $1/1$ and whose labels are Pell numbers, the rule gives successively

$$29 + 2i = (5 + 12i)(1 - 2i),$$
$$169 + 2i = (29 + 12i)(5 - 2i),$$
$$985 + 2i = (29 + 70i)(5 - 12i),$$

that is $169 = 5^2 + 12^2$, $985 = 29^2 + 12^2$ and $5741 = 29^2 + 70^2$, each second factor being the Gaussian integer found two generations earlier. Along the fractions $1/Q$ it recovers the identity $1 + 233i = (21 + 13i)(5 + 8i)$ of the label 610, with both factors multiplied by -1.

Remark 3.9 (The pair determines the occurrence)**.** Two distinct occurrences of one label have distinct occurrence pairs, even up to units and conjugation: equal pairs have equal costs, and Lemma 2.5 applies. Proposition 3.8 computes the occurrence pair from the Farey triangle, that is, from the position of the occurrence in the Markov tree, and not from its label. A rule that produced the occurrence pair from the label alone, correctly for every occurrence, would therefore prove the uniqueness conjecture; Section 7.2 states a conjecture of this kind.

The finite part of the proof, to which we now turn, needs no recursion: it computes the whole fiber of each label, in which the occurrence pair is the one given by the Gram factor of Proposition 2.2.

3.3. **An elementary primality criterion.** The factorizations used below rest on the following classical criterion, which we state for completeness. Suppose $n > 2$, the complete factorization of $n - 1$ into proved primes is supplied, and an integer a satisfies

$$a^{n-1} \equiv 1 \pmod{n}, \qquad \gcd\left(a^{(n-1)/r} - 1, n\right) = 1 \quad \text{for every prime } r \mid n - 1. \tag{21}$$

TABLE 1. The complete finite part of the minimality proof. Each row includes all Q in the displayed interval with $\gcd(P, Q) = 1$. The pair count includes every normalized primitive positive representation of every source, and the last column gives the largest individual fiber. The later sections cover the unbounded denominators beyond these windows.

P	Denominators Q	Sources	Pairs	Largest fiber
8	9–28	10	22	4
9	10–34	17	56	8
10	11–38	11	33	8
11	12–75	59	324	32
12	13–100	29	211	32
13	14–145	122	992	64
Total		248	1638	64

Then n is prime. Indeed, for any prime divisor s of n, these conditions make the order of a modulo s exactly $n - 1$. Since that order divides $s - 1$, one has $s \geqslant n$, hence $s = n$.

Applied recursively, with the primes of $n - 1$ themselves established the same way, the criterion terminates at the prime two and establishes primality by integer arithmetic alone: no probabilistic test and no search for factors enters the verification, only the confirmation of factors already proposed.

Each label is computed twice and the two values are compared: once from the floor increments of the lower Christoffel word, and once from the Cohn matrix built by Stern–Brocot mediants from the two boundary matrices. The occurrence pair is computed from the central Gram factor H of Proposition 2.2. For each label one then enumerates the 2^t sign choices in (14) and passes to the quotient by conjugation; that every normalized pair is obtained exactly twice, as Proposition 3.1 predicts, is itself a check on the enumeration. The Euclidean algorithm then supplies the quotient sequence and the cost of every pair.

3.4. **The finite windows.** The finite part of the proof concerns the reduced fractions listed in Table 1: in row P, for $8 \leqslant P \leqslant 13$, every denominator from $P + 1$ up to the last denominator before the junction Q_P of Section 6. For each of these 248 labels, the prime factorization, the primality of its factors by the recursive use of (21), the splitting of each prime and the costs of the resulting pairs are finitely many assertions about integers. They are recorded as part (i) of the finite verifications, Proposition 6.1, and Proposition 6.3 draws their consequence: in every finite window the occurrence pair is the unique representation of minimum cost.

The statement on the finite windows is stronger than weak minimality, since it also asserts uniqueness of the minimizing representation. The unbounded part requires only weak minimality. It uses the finite windows as exact boundary data and separately excludes every lower-cost competitor on each infinite tail.

4. PATH ENVELOPES AND THE FIRST SEVEN ROWS

The Euclidean bijection turns a cost bound into a finite binary tree. Enumerating that tree at every denominator would obscure the reason that an entire row can be treated. Instead, we compare its branches with a small number of explicit paths. Fibonacci growth controls the cost, and deviations from the Fibonacci path determine the finer norm bands.

4.1. **Sorted paths and a stable order.** On sorted positive pairs define

$$\mathsf{A}(x,y) = (x, x+y), \qquad \mathsf{B}(x,y) = (y, x+y). \tag{22}$$

The first move from $(1,1)$ is symmetric and gives $(1,2)$. Thereafter these two moves are distinct, and every sorted primitive pair of cost $c \geqslant 1$ is obtained by a unique sequence of $c-1$ moves. Indeed, from $0 < x < y$ its sorted predecessor is the sorted pair formed from $x, y-x$. If $y > 2x$, the last move was A; if $y < 2x$, it was B. The equality $y = 2x$ is primitive only at $(1,2)$. This proves the asserted uniqueness directly, including its initial boundary.

A move A after the fixed first move is called a *defect*. Write $v_c(D)$ for the endpoint whose defect costs form $D \subseteq \{2, \dots, c\}$. Thus a defect at cost j is the move arriving at depth j. Products written as words in the moves are read chronologically from left to right; an expression $\mathsf{B}^n z$, with a vector z, means the n-fold application of B to that vector.

Let $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, and use $F_{-n} = (-1)^{n+1} F_n$ when a displayed identity reaches a negative index. Put

$$N(x,y) = x^2 + y^2, \qquad \Lambda(x,y) = x + \varphi y, \qquad \varphi = \frac{1+\sqrt{5}}{2}.$$

Lemma 4.1 (Maximum, replacement, and propagation)**.** *At exact cost c, the unique maximum norm is F_{2c+3}, attained at (F_{c+1}, F_{c+2}). Replacing any defect by B strictly increases the final norm. Finally, simultaneous weak inequalities in N and Λ are preserved by every common B-tail.*

Proof. Coordinate induction in (22), starting from $(1,1)$, gives $x \leqslant F_{c+1}$ and $y \leqslant F_{c+2}$. The identity $F_{c+1}^2 + F_{c+2}^2 = F_{2c+3}$ proves the norm bound. After the first move, $x < y$, and

$$\mathsf{B}(x,y) - \mathsf{A}(x,y) = (y - x, 0).$$

This difference is nonnegative and nonzero. Each subsequent matrix is nonnegative and invertible, so it preserves a nonnegative nonzero difference. Since both endpoint coordinates are positive, the norm strictly increases. Replacing every defect therefore proves uniqueness of the Fibonacci maximum as well. For propagation, direct calculation gives

$$\Lambda(\mathsf{B}z) = \varphi \Lambda(z), \qquad N(\mathsf{B}z) = \varphi^{-2} N(z) + \varphi^{-1} \Lambda(z)^2. \tag{23}$$

Both coefficients are positive, and Λ is positive on the states under consideration. □

Lemma 4.2 (One-defect minimum)**.** *For $c \geqslant 2$, the earliest-defect endpoint $e_c = \mathsf{B}^{c-2}(1,3)$ minimizes both N and Λ among one-defect paths, and*

$$N(e_c) = 5F_{2c-1}. \tag{24}$$

Proof. The assertion starts at $c = 2$. An inherited one-defect path is covered by (23). The only new path creates its defect at the last move and has endpoint $b_c = (F_c, F_{c+2})$. Fibonacci addition gives

$$b_c - e_c = F_{c-2}(-1, 1), \qquad N(b_c) - N(e_c) = 4F_{c-2}F_{c-1}.$$

Its Λ-difference is $(\varphi - 1)F_{c-2}$, so induction closes. The equality for $N(e_c)$ follows by substitution from $e_c = (F_c + F_{c-2}, F_{c+1} + F_{c-1})$, or from the recurrence $N(\mathsf{B}^2 z) = 3N(\mathsf{B}z) - N(z)$ and its first two values. □

4.2. **The second envelope.**

Lemma 4.3 (The second norm envelope)**.** *At cost $c \geqslant 5$, every non-Fibonacci endpoint satisfies*

$$N(x, y) \leqslant E_c := F_c^2 + F_{c+2}^2. \tag{25}$$

Equality is attained at (F_c, F_{c+2}), and also at $(12, 19)$ when $c = 6$; there are no other equality cases.

Proof. Let j be the last defect cost and $r = c - j$. Before that defect, the coordinate bound in Lemma 4.1 gives the upper state (F_j, F_{j+1}). Its final endpoint is

$$X_{c,j} = F_c + F_r F_j, \qquad Y_{c,j} = F_{c+2} - F_r F_{j-1}.$$

Any earlier defect makes at least one final coordinate strictly smaller. Expansion and Cassini's identity give the exact deficit

$$E_c - X_{c,j}^2 - Y_{c,j}^2 = F_r\big[2(F_j F_{j-1} + (-1)^j)F_{r+1} - F_{j-2}F_{j+1}F_r\big]. \tag{26}$$

If $r = 0$, this is zero. For $r \geqslant 1$ and $j \geqslant 4$, replace F_{r+1} by its lower bound F_r; the remaining coefficient is

$$2F_{j-1}^2 - F_{j-2}^2 + 2(-1)^j > 0.$$

The case $j = 2$ is positive directly. When $j = 3$, the bracket in (26) is $2F_{r+1} - 3F_r$. Here $r \geqslant 2$; its values at $r = 2, 3$ are $1, 0$, and for $r \geqslant 4$ it is $F_{r-3} > 0$. The zero cases give exactly the two stated endpoints. □

It follows that an occurrence of cost $L \geqslant 6$ and label m is weakly Euclidean-minimal whenever

$$E_{L-1} < m, \qquad m \neq F_{2L+1}. \tag{27}$$

At cost $L - 1$ only the Fibonacci maximum can lie above E_{L-1}. Every smaller cost has norm at most $F_{2L-1} < E_{L-1}$. Thus (27) excludes every smaller-cost representation.

Figure 3 shows the situation that this criterion exploits. Normalized by the Fibonacci maximum, the norms of all pairs of one cost lie between zero and the second envelope, near 0.829, except for the isolated value 1, so that a hole separates the two. The normalized source labels of a row tend to the constant C_P computed in Section 4.3 below. For $P \leqslant 6$ these limits exceed 1, and C_7 falls in the hole; for $P \geqslant 8$ they fall inside the cloud, where a finer separation is needed, and that is the business of Section 5.

4.3. **The labels of a row.** The source labels of the later rows must be placed in narrow normalized intervals, uniformly in the denominator. This subsection computes the constant C_P that they approach and bounds the approach on both sides, with explicit constants. The idea is to diagonalize the Cohn matrix $\mu(a)$. In its eigenbasis the second Cohn matrix $\mu(b)$ becomes a matrix with positive entries, so that the trace formula (7) turns the label into a sum of positive terms; the term that follows the dominant eigenvector at every letter gives C_P, and the others are corrections whose size is governed by the runs of the letter a.

Put

$$\lambda = \frac{3}{\sqrt{5}\,\varphi} = \frac{15 - 3\sqrt{5}}{10}, \qquad C_P = \varphi^2 \lambda^{P-1}, \qquad \widehat{B} = \frac{1}{\sqrt{5}} \begin{pmatrix} 3\varphi^3 & 2 \\ 2 & 3\varphi^{-3} \end{pmatrix},$$

and, for a reduced fraction P/Q, write its lower Christoffel word as

$$\gamma_{P/Q} = a^{u_1} b\, a^{u_2} b \cdots a^{u_P} b, \qquad u_k \geqslant 0.$$

The normalized label of the occurrence is

$$\rho_{P,Q} = \frac{m_{P/Q}}{F_{2(P+Q-2)+1}}, \tag{28}$$

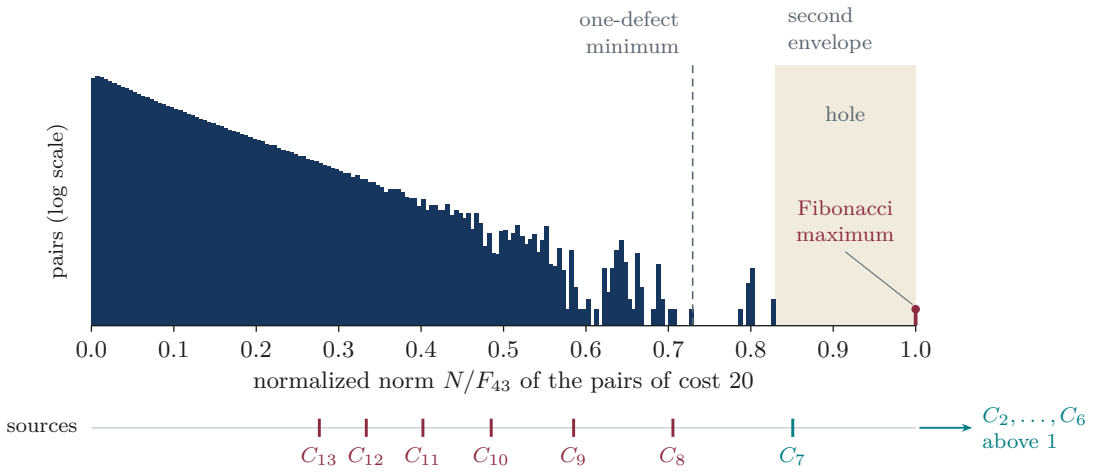


FIGURE 3. The normalized norms N/F_{43} of all 2^{19} sorted primitive pairs of cost 20, counted in bins of width $1/200$ on a logarithmic scale (navy). The Fibonacci maximum is the isolated value 1; every other pair lies at or below the second envelope of Lemma 4.3, and every one-defect pair at or above the minimum of Lemma 4.2 (dashed). The shaded hole contains no normalized norm. The ticks below mark the limits C_P of the normalized source labels of the rows: C_7 lies in the hole, $C_8, \dots, C_{13}$ inside the cloud, and $C_2, \dots, C_6$ above 1.

the label divided by the Fibonacci maximum one cost below the occurrence cost.

Proposition 4.4 (The labels of a row)**.** *Let $0 < P < Q$ be coprime and $n = \lfloor (Q-P)/P \rfloor$.*

(i) *Every run satisfies $n \leqslant u_k \leqslant n+1$, and $u_1 + \cdots + u_P = Q - P$.*

(ii) *The label is the trace*

$$m_{P/Q} = \frac{1}{3} \operatorname{tr} \prod_{k=1}^{P} \begin{pmatrix} \varphi^{2u_k} & 0 \\ 0 & \varphi^{-2u_k} \end{pmatrix} \widehat{B},$$

the factors being multiplied in the order $k = 1, \dots, P$.

(iii) *If $P \geqslant 2$, then*

$$C_P < \rho_{P,Q} < C_P \bigl(1 + \varphi^{-4n-6}\bigr)^P, \tag{29}$$

while $\rho_{1,Q} < C_1$ for the first row. In particular $\rho_{P,Q}$ tends to C_P when Q grows with P fixed.

Proof. The runs. The kth letter b of $\gamma_{P/Q}$ stands at position $\lceil kQ/P \rceil$, as recalled after Proposition 2.2, so that $u_1 = \lceil Q/P \rceil - 1$ and $u_k = \lceil kQ/P \rceil - \lceil (k-1)Q/P \rceil - 1$ for $k \geqslant 2$. For every real x and y the difference $\lceil x+y \rceil - \lceil x \rceil$ is $\lfloor y \rfloor$ or $\lceil y \rceil$; with $y = Q/P$, every u_k is therefore $\lfloor Q/P \rfloor - 1 = n$ or $\lceil Q/P \rceil - 1 \leqslant n+1$. The word has Q letters, of which P are b, so the runs add up to $Q - P$.

The eigenbasis. Let V be the matrix with columns $(\varphi, 1)^{\mathsf{T}}$ and $(1, -\varphi)^{\mathsf{T}}$. Its columns are orthogonal, each of squared length $\varphi^2 + 1 = \sqrt{5}\,\varphi$, so that $O = V/(5^{1/4}\varphi^{1/2})$ is an orthogonal matrix. Since $\varphi^2 = \varphi + 1$,

$$\mu(a) \begin{pmatrix} \varphi \\ 1 \end{pmatrix} = \begin{pmatrix} 2\varphi + 1 \\ \varphi + 1 \end{pmatrix} = \varphi^2 \begin{pmatrix} \varphi \\ 1 \end{pmatrix}, \qquad \mu(a) \begin{pmatrix} 1 \\ -\varphi \end{pmatrix} = \begin{pmatrix} 2 - \varphi \\ 1 - \varphi \end{pmatrix} = \varphi^{-2} \begin{pmatrix} 1 \\ -\varphi \end{pmatrix},$$

because $\varphi^3 = 2\varphi + 1$ and $\varphi^{-2} = 2 - \varphi$. Thus $O^{\mathsf{T}}\mu(a)O = \operatorname{diag}(\varphi^2, \varphi^{-2})$. For the second matrix,

$$V^{\mathsf{T}}\mu(b)V = \begin{pmatrix} 5\varphi^2 + 4\varphi + 1 & 2\varphi \\ 2\varphi & 6 - 3\varphi \end{pmatrix} = \begin{pmatrix} 3\varphi^4 & 2\varphi \\ 2\varphi & 3\varphi^{-2} \end{pmatrix},$$

using $5\varphi^2 + 4\varphi + 1 = 9\varphi + 6 = 3\varphi^4$ and $6 - 3\varphi = 3\varphi^{-2}$; dividing by $\sqrt{5}\,\varphi$ gives $O^{\mathsf{T}}\mu(b)O = \widehat{B}$. All four entries of $\widehat{B}$ are positive, $\operatorname{tr}\widehat{B} = 6$ and $\det\widehat{B} = 1$. Since $\mu(\gamma_{P/Q}) = \prod_k \mu(a)^{u_k}\mu(b)$ and $O^{\mathsf{T}}\mu(a)^u O = \operatorname{diag}(\varphi^{2u}, \varphi^{-2u})$, conjugating by O and taking traces turns (7) into (ii).

Normalization. Write $\beta = \widehat{B}_{12}/\widehat{B}_{11} = 2/(3\varphi^3)$, $\delta = \widehat{B}_{22}/\widehat{B}_{11} = \varphi^{-6}$ and $t_k = \varphi^{-4u_k}$. Taking the factor $\varphi^{2u_k}\widehat{B}_{11}$ out of the kth matrix of (ii) leaves

$$M_k = \begin{pmatrix} 1 & \beta \\ t_k\beta & t_k\delta \end{pmatrix}, \qquad m_{P/Q} = \frac{1}{3}\,\varphi^{2(Q-P)}\Big(\frac{3\varphi^3}{\sqrt{5}}\Big)^P\Theta, \qquad \Theta = \operatorname{tr}\prod_{k=1}^{P} M_k,$$

by (i). At the odd index $2(P+Q)-3$, Binet's formula reads $F_{2(P+Q)-3} = (\varphi^{2(P+Q)-3} + \varphi^{3-2(P+Q)})/\sqrt{5}$. Dividing, and collecting the powers of 3, $\sqrt{5}$ and φ,

$$\rho_{P,Q} = \frac{3^{P-1}\varphi^{3-P}}{5^{(P-1)/2}} \cdot \frac{\Theta}{1+\eta} = C_P\,\frac{\Theta}{1+\eta}, \qquad \eta = \varphi^{6-4(P+Q)}.$$

The lower bound. Expand the trace of the product of the nonnegative matrices M_k as a sum over closed walks $i_1, \dots, i_P, i_{P+1} = i_1$ in $\{1, 2\}$, the walk contributing $\prod_k (M_k)_{i_k i_{k+1}}$. Every contribution is nonnegative. The walk that stays at 1 contributes 1. When $P \geqslant 2$, the walk that visits 2 only at the step $j = 1$ contributes $(M_P)_{12}(M_1)_{21} = \beta^2 t_1$. Hence $\Theta \geqslant 1 + \beta^2 t_1 \geqslant 1 + \beta^2\varphi^{-4(Q-P)}$, since $u_1 \leqslant Q - P$. The ratio of $\beta^2\varphi^{-4(Q-P)}$ to η is

$$\frac{4}{9\varphi^6}\,\varphi^{-4Q+4P}\,\varphi^{4P+4Q-6} = \frac{4}{9}\,\varphi^{8P-12} \geqslant \frac{4}{9}\,\varphi^4 > 1,$$

because $P \geqslant 2$ and $\varphi^4 = (7 + 3\sqrt{5})/2 > 9/4$. Therefore $\Theta > 1 + \eta$ and $\rho_{P,Q} > C_P$.

The upper bound. By (i), $0 \leqslant t_k \leqslant \tau := \varphi^{-4n}$, so that each M_k is bounded entrywise by $M = \left(\begin{smallmatrix} 1 & \beta \\ \tau\beta & \tau\delta \end{smallmatrix}\right)$, and since products of nonnegative matrices are monotone entrywise, $\Theta \leqslant \operatorname{tr} M^P$. The matrix M has trace $1 + \tau\delta > 0$ and determinant $\tau(\delta - \beta^2) = 5\tau/(9\varphi^6) > 0$; its discriminant $(1 + \tau\delta)^2 - \frac{20}{9}\tau\delta = 1 - \frac{2}{9}\tau\delta + (\tau\delta)^2$ is positive because $\tau\delta \leqslant \varphi^{-6} < 1$. Its eigenvalues μ_1, μ_2 are therefore positive, and

$$\operatorname{tr} M^P = \mu_1^P + \mu_2^P \leqslant (\mu_1 + \mu_2)^P = (1 + \tau\delta)^P = \big(1 + \varphi^{-4n-6}\big)^P,$$

the binomial expansion of the right-hand side containing the left-hand side among nonnegative terms. As $1 + \eta > 1$, the upper bound in (29) follows.

The first row. When $P = 1$ the product has the single factor M_1 and $u_1 = Q - 1$, so that $\Theta = 1 + t_1\delta = 1 + \varphi^{-4Q-2}$, which is smaller than $1 + \eta = 1 + \varphi^{2-4Q}$. Hence $\rho_{1,Q} < C_1$, and $\rho_{1,Q} = C_1(1 + \varphi^{-4Q-2})/(1 + \varphi^{2-4Q})$ tends to C_1. Finally, when $P \geqslant 2$, the integer $n \geqslant (Q - 2P + 1)/P$ grows without bound with Q, and both sides of (29) tend to C_P. □

The proposition shows where the constant comes from. In the leading term of (ii), each letter a contributes the factor φ^2 and each letter b the factor $\widehat{B}_{11} = 3\varphi^3/\sqrt{5} = \lambda\varphi^4$; the Fibonacci normalization grows by φ^2 for each unit of cost, a letter a adding one unit and a letter b two. Every letter b therefore costs exactly the factor λ against the Fibonacci growth, and $C_P = (\varphi^2/\lambda)\lambda^P$. The same number appears on the side of the paths. Normalized by the maximum at its own cost, the second envelope of Lemma 4.3 tends to λ, since $E_c/F_{2c+3} = (F_c^2 + F_{c+2}^2)/(F_{c+1}^2 + F_{c+2}^2)$ and $1 + \varphi^4 = 3\varphi^2$; and the second row lies on that envelope, because for odd $Q = 2n + 3$ the word $\gamma_{2/Q}$ has central

factor $a^n b a^n$, whose Gram factor $P_2\,\mu(a)^n$ sends $e = (F_3, F_2)^{\mathsf{T}}$ to $(F_{Q+2}, F_Q)^{\mathsf{T}}$, so that $m_{2/Q} = E_Q$. The value λ is also the limit of the one-defect ratio of Figure 4(a).

Lee, Li, Rabideau and Schiffler prove, in their coordinates, that for fixed $P \geqslant 2$ the ratio (28) does not increase with Q [16, Lemma 6.4], and they compute its limit in the proof of their Lemma 6.6 [16, equation (6.4)].[2] Neither result is used below: Proposition 4.4 gives the lower bound for every denominator directly, and its upper bound, completed by finitely many exact comparisons, replaces the monotonicity in Section 6.

4.4. **The first seven rows.** With the second envelope and the labels of a row at hand, the first seven rows follow from norms alone: their source labels sit above every norm of smaller cost except possibly the Fibonacci maximum, and that maximum is avoided.

Proposition 4.5. *Every occurrence with $1 \leqslant P \leqslant 7$ is weakly Euclidean-minimal.*

Proof. The first row has $m_{1/Q} = F_{2Q+1}$ and cost $L = Q-1$. Every smaller cost has norm at most F_{2Q-1}, proving this case. The row formula follows, for example, by iterating the first Cohn matrix in (7).

For $L \geqslant 6$ put $u = F_{L-1}/F_L$. The interval $3/5 \leqslant u \leqslant 5/8$ is preserved by $u \mapsto 1/(1+u)$. The inequality $E_{L-1} < (5/6)F_{2L+1}$ reduces to $7u^2 + 2u - 4 < 0$, whose maximum on this interval is $-1/64$. The constants C_P decrease with P, and

$$C_7 = \frac{729}{125\varphi^4} > \frac{5}{6}, \qquad C_6 = \frac{243}{25\sqrt{5}\varphi^3} > 1.$$

The two radical signs reduce respectively to $4373 > 1875\sqrt{5}$ and $59 > 25\sqrt{5}$, with squared margins 1545004 and 356. Hence (29) proves (27) for $2 \leqslant P \leqslant 6$. For $P = 7$, the lower bound of (29) and $C_7 > 5/6$ give $m > E_{L-1}$, and it remains to show that $m \neq F_{2L+1}$, that is $\rho_{7,Q} \neq 1$; we show $\rho_{7,Q} < 1$. When $Q \geqslant 14$, every run of $\gamma_{7/Q}$ has length at least $\lfloor (Q-7)/7 \rfloor \geqslant 1$, and Proposition 4.4 gives $\rho_{7,Q} < C_7(1+\varphi^{-10})^7$. Since $\varphi^{10} = (123+55\sqrt{5})/2 > 120$ and $\varphi^4 = (7+3\sqrt{5})/2 > 6.85$, one has $(1+\varphi^{-10})^7 < e^{7/120} < e^{0.06} < 1.07$ and $C_7 = 729/(125\varphi^4) < 729/856 < 0.86$, so that $\rho_{7,Q} < 0.86 \cdot 1.07 < 0.93$. For $8 \leqslant Q \leqslant 13$ the six labels 195025, 499393, 1278818, 3276509, 8399329 and 21531778 lie below $F_{27} = 196418$, $F_{29} = 514229$, $F_{31} = 1346269$, $F_{33} = 3524578$, $F_{35} = 9227465$ and $F_{37} = 24157817$ respectively. Thus this row also avoids the Fibonacci hole.

The remaining $L \leqslant 5$ slopes are $1/2, 1/3, 1/4, 1/5, 1/6, 2/3, 2/5, 3/4$. Their (L, m) values are respectively

$$(1,5), (2,13), (3,34), (4,89), (5,233), (3,29), (5,194), (5,169).$$

Each satisfies $m > F_{2L+1}$, the maximum at costs at most $L-1$. □

The later rows require a finer separation than the second envelope. The next section classifies fixed-defect paths by their gap parities, and Section 6 combines those unbounded covers with the complete finite Gaussian fibers of Section 3.

5. Covering the unbounded tails

A finite denominator check cannot establish weak minimality along an entire Farey row. The missing step is a uniform separation between the source ratio and every shorter inverse-Euclidean path. We obtain this separation by covering an infinite lattice of gap

[2] Two printed slips of that proof should be corrected. The equality labels the first-row value $m_{q,1}$ as F_{2q-1} on the line that carries the Binet expression of F_{2q+1}, which is the correct index. And the monotonicity is asserted there for every positive parameter, whereas at the parameter of the first row the ratio is F_{2Q+1}/F_{2Q-1}, which increases strictly to φ^2 and stays below its limit, as Proposition 4.4(iii) also shows.

lengths with finitely many boxes. On each box the normalized ratio is controlled by its two values at a matched pair of corners, and the reason two values suffice is an order theorem for that ratio, Fibonacci denominator included.

5.1. **A common eigenchannel for every defect count.** Use the inverse-Euclidean matrices

$$\mathsf{A} = \begin{pmatrix} 1 & 0 \\ 1 & 1 \end{pmatrix}, \qquad \mathsf{B} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \qquad z_0 = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

For $k \geqslant 1$, a path with exactly k defects is described uniquely by nonnegative gaps $g = (g_0, \dots, g_k)$: apply B^{g_i} and then A for $i < k$, and apply only B^{g_k} at the end. The order in this description is chronological. If $S = \sum_i g_i$, the cost is $c = S + k + 1$.

Put $\varphi = (1 + \sqrt{5})/2$, $\theta = \varphi^{-2}$, and $q = -\theta = \varphi - 2$. The matrix

$$T = \begin{pmatrix} 1 & \varphi \\ \varphi & -1 \end{pmatrix}$$

satisfies $T^2 = (1 + \varphi^2)I$. Moreover

$$\varphi^{-g} T \mathsf{B}^g T^{-1} = \operatorname{diag}(1, q^g), \qquad \sqrt{5}\varphi^{-1} T \mathsf{A} T^{-1} = \begin{pmatrix} 2 & 1 \\ q & -2q \end{pmatrix}.$$

Thus we may start with $(X, Y) = (\varphi^2, 2 - \sqrt{5})$, apply

$$\begin{pmatrix} X \\ Y \end{pmatrix} \longmapsto \begin{pmatrix} 2 & q^{g_i} \\ q & -2q^{g_i+1} \end{pmatrix} \begin{pmatrix} X \\ Y \end{pmatrix} \quad (i < k), \qquad (X, Y) \longmapsto (X, q^{g_k} Y) \quad (i = k), \tag{30}$$

and denote the final $X^2 + Y^2$ by $\Pi_k(g)$. The initial vector is $\varphi^{-1} T z_0$, so the product of all removed positive factors gives

$$\Pi_k(g) = 5^k \varphi^{-2c} (1 + \varphi^2) N(g).$$

Binet's formula, with $2c + 3$ odd, consequently gives

$$\mathcal{R}_k(g) := \frac{N(g)}{F_{2c+3}} = \frac{\Pi_k(g)}{K_k (1 - q^{2k+5+2S})}, \qquad K_k = 5^k \varphi^4. \tag{31}$$

The denominator is positive. This formula includes zero gaps and the special final gap, which has no following defect matrix.

The right-hand side of (31) depends on the gaps only through the numbers $x_j = q^{g_j}$ for $j < k$, which enter the matrices (30), and $y = \theta^{2g_k}$, since $q^{2S} = y \prod_{j<k} x_j^2$. It is a rational function of these variables whose denominator stays positive when any of them is set to zero. We use it to give a meaning to infinite gaps: the value at $g_j = \infty$ is the value at $x_j = 0$, or at $y = 0$ for the final gap. An even gap has $x_j \in (0, 1]$, an odd gap has $x_j \in [-\theta, 0)$, and the final gap has $y \in (0, 1]$; with the infinite values added, the ranges become the closed intervals $[0, 1]$, $[-\theta, 0]$ and $[0, 1]$.

5.2. **Order in each parity coordinate.**

Theorem 5.1 (Normalized parity order)**.** *For every $k \geqslant 1$, hold all gaps but g_j fixed and increase g_j by two. The normalized ratio $\mathcal{R}_k$ strictly increases if $j < k$ and g_j is even. It strictly decreases if $j < k$ and g_j is odd, or if $j = k$ with either parity. More precisely, on the closed ranges just described, $\mathcal{R}_k$ is a strictly decreasing function of each x_j, $j < k$, and a strictly increasing function of y, whatever the values of the other variables in their closed ranges; in particular the monotonicity persists when some of the other gaps are infinite.*

Proof. The state before a gap. Just before any gap, the physical column has second-to-first coordinate ratio $\xi \geqslant 2$. This is true at $(1,2)$; every intervening B block leaves a ratio greater than one, and the following A adds one. Its channel therefore has $X > 0$, $Y < 0$, and

$$h = -Y/X = \frac{\xi - \varphi}{1 + \varphi\xi}, \qquad \varphi^{-5} \leqslant h < \varphi^{-1}.$$

This includes the first gap.

A nonfinal gap. Write the two columns of the channel defect matrix as $\mathsf{u} = (2,q)^{\mathsf{T}}$ and $\mathsf{v} = (1,-2q)^{\mathsf{T}}$. Their physical preimages are

$$T^{-1}\mathsf{u} = \begin{pmatrix} \varphi^{-2} \\ 1 \end{pmatrix} =: \mathsf{a}, \qquad T^{-1}\mathsf{v} = \begin{pmatrix} \varphi^{-1} \\ \varphi^{-3} \end{pmatrix} =: \mathsf{b}, \qquad \varphi^{-3}\mathsf{a} \leqslant \mathsf{b} \leqslant \varphi\mathsf{a}.$$

Let S be the product of the physical matrices that follow the defect, a matrix with nonnegative entries, and put $U = S\mathsf{a}$ and $V = S\mathsf{b}$. Componentwise $\varphi^{-3}U \leqslant V \leqslant \varphi U$, all entries being nonnegative, so that

$$\Delta := \langle U, V \rangle \geqslant \varphi^{-3}\|U\|^2 > 0, \qquad \nu := \|V\|^2 \leqslant \varphi\Delta;$$

here $U \neq 0$ because every factor of S maps positive columns to positive columns. The gap and the defect send the channel vector $(X, -hX)$ to $X(\mathsf{u} - xh\mathsf{v})$, with $x = x_j = q^{g_j}$, and the channel matrices that follow are positive multiples of TST^{-1}. Since T is scaled orthogonal, $T^{\mathsf{T}}T = (1+\varphi^2)I$, the numerator is therefore a positive multiple, independent of x, of

$$\Pi(x) = X^2(\|U\|^2 - 2xh\Delta + x^2h^2\nu).$$

An even gap. Here $0 \leqslant x \leqslant 1$, and $xh\nu < \Delta$ on this whole closed range, because $h\nu < \varphi^{-1}\varphi\Delta = \Delta$. The derivative $2X^2h(xh\nu - \Delta)$ of Π is therefore negative, and Π decreases strictly with x. The normalizing denominator $1 - q^{2k+5+2S} = 1 + \theta^{2k+5}\prod_i x_i^2 \cdot y$ increases with x^2, so $\mathcal{R}_k$ decreases strictly with x on $[0,1]$. Increasing an even gap by two decreases x, and so increases $\mathcal{R}_k$. This proves the even case.

An odd gap. Here $x = -\tau$ with $0 \leqslant \tau \leqslant \theta$. Set $\zeta = \theta^{2k+5+2\sum_{i\neq j} g_i}$, so that the normalizing denominator is $1 + \zeta\tau^2$. Up to the fixed positive factor K_k, the normalized function is

$$F(\tau) = \frac{\omega_0 + 2\omega_1\tau + \omega_2\tau^2}{1 + \zeta\tau^2}, \qquad \omega_0 = X^2\|U\|^2, \quad \omega_1 = X^2h\Delta, \quad \omega_2 = X^2h^2\nu.$$

From the bounds on Δ and h, $\omega_0/\omega_1 \leqslant \varphi^8 = \theta^{-4}$. Differentiation gives

$$F'(\tau) = \frac{2(\omega_1 + \omega_2\tau - \omega_0\zeta\tau - \omega_1\zeta\tau^2)}{(1 + \zeta\tau^2)^2},$$

and the numerator is positive on the closed range: using $\omega_2\tau \geqslant 0$, $\tau \leqslant \theta$ and $\zeta \leqslant \theta^{2k+5}$,

$$\omega_1 + \omega_2\tau - \omega_0\zeta\tau - \omega_1\zeta\tau^2 \geqslant \omega_1\big(1 - \theta^{-4}\theta^{2k+5}\theta - \theta^{2k+5}\theta^2\big) = \omega_1\big(1 - \theta^{2k+2} - \theta^{2k+7}\big) > 0.$$

Indeed $\theta < 1/2$ and $k \geqslant 1$, so even the boundary case has $\theta^4 + \theta^9 < 1$. Thus F increases strictly with τ, that is, $\mathcal{R}_k$ decreases strictly with $x = -\tau$ on $[-\theta, 0]$. Increasing the odd gap by two decreases τ, and therefore decreases $\mathcal{R}_k$.

The final gap. Put $\zeta = \theta^{2k+5+2\sum_{i<k} g_i}$. Up to K_k, the normalized function of $y = \theta^{2g_k}$ is

$$\frac{X^2 + Y^2 y}{1 + \zeta y},$$

whose derivative in y is $(Y^2 - X^2\zeta)/(1+\zeta y)^2$, of the sign of $h^2 - \zeta$. This is positive since $h^2 \geqslant \theta^5 > \theta^{2k+5} \geqslant \zeta$, including $k = 1$. So $\mathcal{R}_k$ increases strictly with y on $[0,1]$; increasing the final gap decreases y and proves the last assertion.

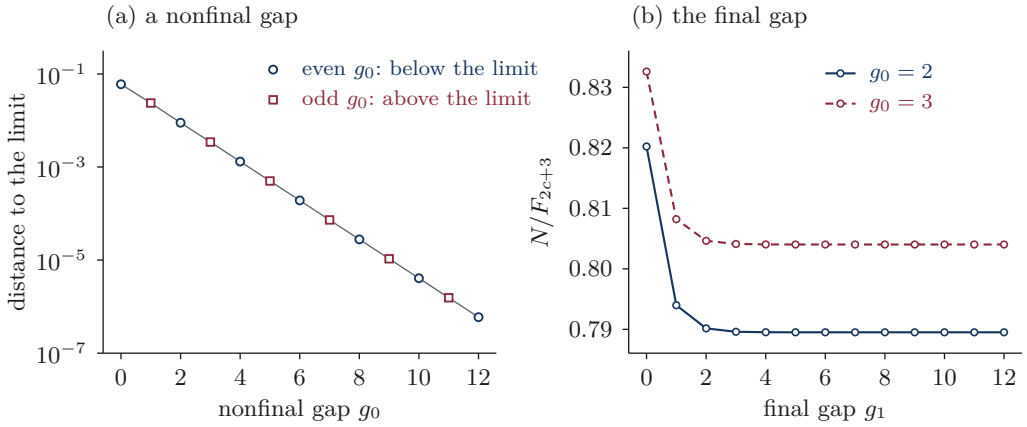


FIGURE 4. The normalized parity order for one defect, computed exactly. (a) The one-defect paths with final gap 0, as a function of the gap g_0 before the defect: the distance of the ratio N/F_{2c+3} from its value $3/(\sqrt{5}\,\varphi) = (15 - 3\sqrt{5})/10$ at $g_0 = \infty$, on a logarithmic axis. The even values (navy circles) lie below that limit and increase towards it, the odd values (wine squares) lie above it and decrease. (b) The ratio itself as a function of the final gap g_1, for $g_0 = 2$ and $g_0 = 3$: it decreases for either parity.

The other variables anywhere in their ranges. The three derivative computations used only the bounds on h, Δ, ν and ζ, and these hold whatever the values of the other variables in their closed ranges, not only at the values q^g, θ^{2g} and 0. Indeed, for $-\theta \leqslant x \leqslant 1$ the channel action $\operatorname{diag}(1, x)$ of a gap corresponds in physical coordinates to

$$M_x = T^{-1} \operatorname{diag}(1, x)\, T = \frac{1}{1+\varphi^2} \begin{pmatrix} 1+\varphi^2 x & \varphi(1-x) \\ \varphi(1-x) & \varphi^2 + x \end{pmatrix},$$

whose entries are nonnegative because $x \geqslant -\theta = -\varphi^{-2}$ and $x \leqslant 1$; it is $\varphi^{-g}\mathsf{B}^g$ when $x = q^g$, and for $x = 0$ it is the limit that represents an infinite gap. It maps positive columns to positive columns. The final gap enters only through y, and acts on the channel as $\operatorname{diag}(1, \sqrt{y})$ with $0 \leqslant \sqrt{y} \leqslant 1$, a matrix of the same kind. Every suffix is thus still a product of nonnegative matrices, and the bounds on Δ and ν hold as shown above. Next, applied to a column $(1, \xi)^{\mathsf{T}}$ with $\xi \geqslant 2$, the matrix M_x gives a column with positive entries whose second coordinate exceeds the first by

$$\frac{\varphi^{-1} + \xi + x\varphi^2(\xi - \varphi)}{1+\varphi^2} \geqslant \frac{\varphi^{-1} + \varphi}{1+\varphi^2} > 0,$$

since for $x < 0$ one has $x\varphi^2(\xi - \varphi) \geqslant -\theta\varphi^2(\xi - \varphi) = -(\xi - \varphi)$; the next A therefore gives a ratio above 2 again, and the bounds on h hold before every gap. Finally, every other variable has absolute value at most 1, so $\zeta \leqslant \theta^{2k+5}$ in both places where it was used. Each of the three derivative computations therefore applies at every point of the closed ranges, and the monotonicity is strict there. With $x_i = 0$ or $y = 0$ this is the case of infinite gaps, which the covers below use. □

Figure 4 shows the theorem at work in the simplest stratum, one defect, where the ratio depends on two gaps only: along a nonfinal gap the even and the odd values form two monotone sequences that approach the common limit from opposite sides, and along the final gap the ratio decreases.

5.3. **A finite cover of the infinite gap lattice.** Fix the parity of every gap. A coordinate domain is either a finite step-two interval or an infinite step-two tail. For a product box, its lower matched corner uses the lower endpoint in increasing coordinates and the upper endpoint in decreasing coordinates; the upper matched corner makes the opposite choices. An infinite endpoint means $x_j = 0$, or $y = 0$ for the final gap. The numerator and denominator in (31) are always evaluated at the same corner. Moving one coordinate at a time from a point of the box to a corner, and applying Theorem 5.1 on the closed ranges at each move, gives

$$\mathcal{R}_k(g^{\min}) \leqslant \mathcal{R}_k(g) \leqslant \mathcal{R}_k(g^{\max}). \tag{32}$$

A corner need not satisfy a subsequently imposed lower bound on the total gap sum; it still safely bounds that restricted part of the box.

For a required total $S \geqslant S_0$, set $\eta = \lceil S_0/(k+1) \rceil$. Every eligible tuple has some gap at least η. Partition it by the first such gap: earlier coordinates are below η, that coordinate is at least η, and later coordinates are unrestricted. Intersecting with a fixed parity gives $k+1$ disjoint root boxes whenever $\eta \geqslant 2$. Empty parity intersections are discarded. Their union contains every eligible tuple exactly once.

Every nonterminal box is split into two disjoint step-two domains along one coordinate, namely the non-singleton coordinate whose domain has the smallest lower endpoint, the first such coordinate in case of a tie: a finite domain of n points is cut into its first $\lfloor n/2 \rfloor$ points and the rest, and an infinite tail into its first point and the rest. A finite box may be discarded if its maximum total is less than S_0. Otherwise it closes only when its upper matched corner is strictly below the lower source threshold, or its lower matched corner is strictly above the upper source threshold. A singleton inside the closed source band would be an explicit failure. A subdivision tree all of whose leaves close therefore settles the whole infinite domain: the root boxes cover the eligible tuples, each exactly once, each split partitions a box, and every leaf lies entirely on one side of the band by (32). Whether a given tree closes is a finite question, settled for the trees of this paper in Proposition 6.1.

All the quantities compared here lie in $\mathbb{Z}[\varphi]$: the channel entries, every power of q, and K_k. Writing $a + b\varphi$ for the pair of integers (a, b) and clearing threshold denominators beforehand, the sign of $a + b\varphi$ is the sign of $2a + b + b\sqrt{5}$, which is decided by the signs of the two integers $2a + b$ and b and, when these oppose each other, by the sign of $(2a + b)^2 - 5b^2$. Every comparison below is therefore an exact comparison of integers; no decimal approximation of $\sqrt{5}$ enters.

5.4. **The six unbounded source windows.** With the constant $C_P = \varphi^2 \lambda^{P-1}$ of Section 4.3, which the source labels of row P approach, define the rational ceilings and starting costs by

P	c_P	U_P	middle defects
8	34	707/1000	2
9	41	2929/5000	2, 3
10	46	4857/10000	2, 3, 4
11	84	20111/50000	2, 3, 4, 5
12	110	$333511/10^6$	2, 3, 4, 5, 6
13	156	$2765404267/10^{10}$	2, 3, 4, 5, 6, 7.

(33)

For row P, the closed source band is $[C_P, U_P]$. The paths with no defect and with a single defect are excluded directly in Section 6, by the Fibonacci maximum and by Lemma 4.2; the paths with at least two defects are the business of the covers. For each pair (P, k) with $2 \leqslant k \leqslant P - 6$ (the middle defects of the table) the cover is built with

$S_0 = c_P - k - 1$ and both thresholds; for $k = P - 5$ with the lower threshold alone. Theorem 6.4 in Section 6 states what the closed trees prove.

The covers prove only the source-band exclusions needed at the actual tail junctions. They do not classify every extremal ray or assert the same bands at smaller costs. That narrower target makes one uniform order theorem sufficient for every stratum.

6. Finite verifications and the proof of the theorem

We now prove the main theorem. Its computer-assisted part consists of finitely many assertions about integers and about elements of $\mathbb{Z}[\varphi]$; we state them together, derive from them the three statements on which the assembly rests, and then join the finite windows to the infinite tails at explicit denominators, so that no interval between a finite calculation and an asymptotic statement is left untreated. Throughout this section, for $P = 8, \dots, 13$ respectively, put

$$Q_P = 29, 35, 39, 76, 101, 146. \tag{34}$$

These are one beyond the six finite windows of Table 1. They satisfy $P + Q_P - 3 = c_P$, with c_P as in (33), and all six fractions P/Q_P are reduced. Beyond a second denominator the upper bound of Proposition 4.4 alone keeps a source below its ceiling U_P; with

$$n_P = 3, 4, 4, 7, 8, 11, \qquad Q_P^* = P(n_P + 1) = 32, 45, 50, 88, 108, 156, \tag{35}$$

every reduced P/Q with $Q \geqslant Q_P^*$ has runs of length at least n_P, and the denominators $Q_P \leqslant Q < Q_P^*$ in between are finitely many.

6.1. The finite verifications.

Proposition 6.1 (Finite verifications)**.** *The following assertions hold.*

(i) *For each of the* 248 *reduced fractions P/Q of Table 1, the label $m_{P/Q}$ is a product of explicit prime powers; each of these primes passes the criterion* (21) *with explicit witnesses, recursively down to the prime two; each odd one is congruent to* 1 *modulo* 4 *and has an explicit decomposition as a sum of two squares; and among the pairs* (14) *built from these data, the occurrence pair of Proposition 2.2 has cost $P+Q-2$ and every other pair has a larger cost.*

(ii) *For each $P \in \{8, \dots, 13\}$ and each k with $2 \leqslant k \leqslant P - 6$, the subdivision tree of Section 5 built from the root boxes of all 2^{k+1} parity sectors, with $S_0 = c_P - k - 1$ and the thresholds C_P and U_P, is finite, and each of its leaves closes: its upper matched corner is strictly below C_P or its lower matched corner is strictly above U_P. For $k = P - 5$ and $S_0 = c_P - k - 1$, every root box has its upper matched corner strictly below C_P.*

(iii) *For $k = 2$ and $S_0 = 152$, every root box of the eight parity sectors has its upper matched corner strictly below* $71/100$.

(iv) *For each $P \in \{8, \dots, 13\}$, one has $C_P(1 + \varphi^{-4n_P-6})^P < U_P$, and $m_{P/Q} < U_P F_{2(P+Q-2)+1}$ for each of the reduced Q with $Q_P \leqslant Q < Q_P^*$, thirty-eight denominators in all.*

Proof. Each assertion is a finite statement about integers and about elements of $\mathbb{Z}[\varphi]$, whose signs are decided exactly by the integer test of Section 5. They were verified by computer in exact arithmetic, and the verification was carried out twice independently, the two computations resting on different formulas. For part (i), the first computation builds the fiber of each label from the sign cube (14) and proves every prime by the criterion (21), applied recursively down to the prime two; the second finds every representation afresh, from the square roots of -1 modulo the label, combined by the Chinese remainder theorem and reduced by Cornacchia's descent, and checks every prime again, by the same

criterion with independently written arithmetic and by a probable-prime test. In both, each label is computed from its Cohn matrix, the first computation obtaining it in two ways, from the floor increments of the Christoffel word and by Stern–Brocot mediants. For parts (ii) and (iii), the first computation evaluates the normalized ratio through the channel recursion (30) in $\mathbb{Z}[\varphi]$; the second rebuilds every tree from its root boxes and evaluates the ratio through the spectral projectors of the matrix B in $\mathbb{Z}[\sqrt{5}]$. For part (iv), the first computation takes each label from its Cohn matrix and decides the constant comparison by the integer sign test in $\mathbb{Q}(\sqrt{5})$; the second takes each label as the continuant of the palindromic continued fraction $[2, \sigma(z), 2]$ of Çanakçı and Schiffler, the Fibonacci numbers from the doubling formulas, and bounds the constants with rational enclosures of $\sqrt{5}$ to sixty decimal places. □

For instance, the first integer comparison of part (iv) is the positive integer identity

$$707F_{71} - 1000\, m_{8/29} = 25054682255203.$$

Remark 6.2. The computer contributes only the assertions of Proposition 6.1: the factorizations and the costs of 1638 explicit pairs, the closing of the boxes of finitely many trees, thirty-eight comparisons of integers and six comparisons of constants. Everything that turns them into statements about infinitely many denominators is proved in the text: that the sign cube lists the whole fiber (Proposition 3.1), that two matched corners bound a whole box (Theorem 5.1 and (32)), that the root boxes cover every admissible gap tuple exactly once and each split partitions a box (Section 5), that a source row stays in its band beyond the junction (Proposition 4.4 and (35)), and the comparisons at the remaining costs, below.

6.2. **Three consequences.**

Proposition 6.3 (Complete finite windows)**.** *For every reduced P/Q in Table 1, the occurrence pair is the unique element of $\mathcal{F}(m_{P/Q})$ with minimum Euclidean cost. That cost is $P + Q - 2$.*

Proof. By Proposition 6.1(i) the label is a product of proved primes, the odd ones split as recorded. By Proposition 3.1, the pairs (14) built from this factorization are the whole fiber $\mathcal{F}(m_{P/Q})$. By Proposition 6.1(i) again, the occurrence pair is the only one of them with cost $P + Q - 2$, and all others cost more. The table contains every reduced denominator of each window, so no occurrence of the windows is missing. □

Theorem 6.4 (Uniform tail exclusion)**.** *For each $P \in \{8, 9, 10, 11, 12, 13\}$, each $2 \leqslant k \leqslant P - 6$, and every nonnegative gap tuple with cost $c \geqslant c_P$, one has*

$$\frac{N(g)}{F_{2c+3}} < C_P \quad \text{or} \quad \frac{N(g)}{F_{2c+3}} > U_P.$$

The inequalities are strict, so no endpoint in these strata belongs to the closed source band $[C_P, U_P]$. Every path with $k \geqslant P - 5$ defects and cost $c \geqslant c_P$ satisfies the stronger conclusion $N(g)/F_{2c+3} < C_P$.

Proof. Middle defect counts. Fix P and k with $2 \leqslant k \leqslant P - 6$. A tuple with k defects has cost $c = S + k + 1$, so $c \geqslant c_P$ means $S \geqslant S_0 = c_P - k - 1$. By Section 5, the root boxes of all parity sectors cover the tuples with $S \geqslant S_0$, each exactly once, and each split partitions a box into two. By Proposition 6.1(ii) the resulting tree is finite and every leaf closes, so each admissible tuple g lies in exactly one closed leaf. If the upper matched corner of that leaf is below C_P, then (32) gives $\mathcal{R}_k(g) \leqslant \mathcal{R}_k(g^{\max}) < C_P$; otherwise its lower matched corner is above U_P and $\mathcal{R}_k(g) \geqslant \mathcal{R}_k(g^{\min}) > U_P$.

The defect count $P-5$. By Proposition 6.1(ii) every root box has its upper matched corner below C_P, and (32) gives $\mathcal{R}_{P-5}(g) < C_P$ on the whole root box, hence for every admissible tuple.

More defects. If a path has more than $P-5$ defects, replace surplus A moves by B as in Lemma 4.1. At an ordered positive state (u,v) the replacement increases the column by $(v-u,0)^{\mathsf{T}}$, and every subsequent matrix is nonnegative and invertible. The final norm strictly increases, while the cost stays fixed. Replacing until exactly $P-5$ defects remain gives a path of the same cost whose norm exceeds the original one and is below $C_P F_{2c+3}$. □

Corollary 6.5 (A lower-cost bound)**.** *Every path of cost* $c \geqslant 155$ *with at least two defects satisfies*

$$\frac{N(g)}{F_{2c+3}} < \frac{71}{100}.$$

Proof. A path with exactly two defects and cost $c \geqslant 155$ has $S \geqslant 152$, so it lies in one root box of the cover with $S_0 = 152$. By Proposition 6.1(iii) and (32), its normalized ratio is below the value at the upper matched corner of that box, hence below $71/100$. A path with more defects is handled by replacing surplus defects, as in the proof of Theorem 6.4. □

The corollary is used only at the cost-$(L-2)$ interface in row thirteen.

6.3. **Where the source intervals meet the covers.** Put $L = P+Q-2$ and $c = L-1$. The occurrence pair has cost L, and we must exclude every representation of the same norm at every smaller cost. We first place the source in its band: for every reduced P/Q with $Q \geqslant Q_P$,

$$C_P < \frac{m_{P/Q}}{F_{2c+3}} < U_P. \tag{36}$$

The lower bound is Proposition 4.4(iii), since $F_{2c+3} = F_{2(P+Q-2)+1}$. For the upper bound there are two cases. If $Q \geqslant Q_P^*$, every run of $\gamma_{P/Q}$ has length at least $n = \lfloor (Q-P)/P \rfloor \geqslant n_P$, and since the upper bound of Proposition 4.4(iii) decreases with n,

$$\frac{m_{P/Q}}{F_{2c+3}} < C_P\bigl(1+\varphi^{-4n_P-6}\bigr)^P < U_P, \tag{37}$$

the second inequality being part (iv) of Proposition 6.1. If $Q_P \leqslant Q < Q_P^*$, the upper bound is one of the thirty-eight integer comparisons of the same part. Figure 5 shows the band, the source labels and the bound of Proposition 4.4 for the row of numerator eight.

6.4. **Excluding cost one step below the occurrence.** The path with no defect has norm F_{2c+3}, above every source ceiling. A one-defect path satisfies

$$\frac{N}{F_{2c+3}} \geqslant \frac{5F_{2c-1}}{F_{2c+3}} > 5\varphi^{-4} > \frac{29}{40} > U_8 \geqslant U_P.$$

The first inequality is Lemma 4.2; the strict Fibonacci comparison follows from Binet's formula at odd indices. Theorem 6.4 excludes the closed source band for every remaining defect count. Thus no representation at cost $L-1$ has norm $m_{P/Q}$.

6.5. **Excluding all smaller costs.** It remains necessary to check smaller costs: exclusion at one depth alone would not imply minimality. We use the following elementary consequences of Binet's formula. For positive odd n, the ratios F_n/F_{n+2} decrease strictly to φ^{-2}, and F_n/F_{n+4} decrease strictly to φ^{-4}. In particular, for odd $n \geqslant 5$,

$$\varphi^{-2} < \frac{F_n}{F_{n+2}} \leqslant \frac{5}{13} < \frac{2}{5}, \qquad \frac{F_n}{F_{n+4}} \leqslant \frac{5}{34} < \frac{3}{20}. \tag{38}$$

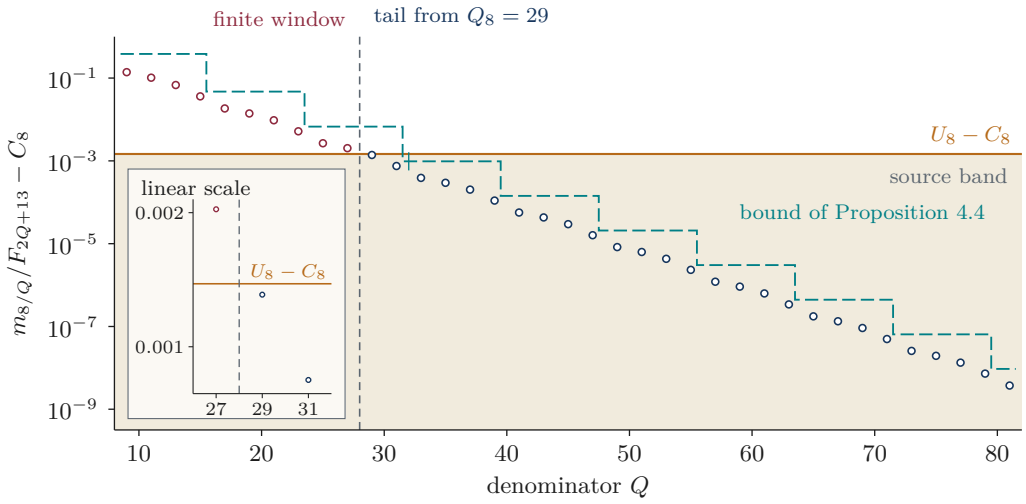


FIGURE 5. The source band of the row $P = 8$. The normalized labels $m_{8/Q}/F_{2(Q+6)+1}$ (navy circles) decrease with the denominator and tend to C_8; from the junction $Q_8 = 29$ on they lie in the band between C_8 and $U_8 = 707/1000$ (shaded), which contains no normalized norm of a path one step cheaper: Theorem 6.4 excludes the paths with at least two defects, and the paths with fewer are excluded at the start of Section 6.4. Before the junction, the wine circles are the labels of the finite window, settled by their complete Gaussian fibers. The dashed staircase is the upper bound of Proposition 4.4, $C_8((1+\varphi^{-4n-6})^8 - 1)$ with $n = \lfloor (Q-8)/8 \rfloor$; it lies above every label and enters the band at $Q_8^* = 32$, so that only the denominators 29 and 31 need a direct comparison. The vertical scale is $(m_{8/Q}/F_{2(Q+6)+1} - C_8)$, on a logarithmic axis. The inset repeats the denominators 27, 29 and 31 on a linear scale: at the junction the label lies inside the band by 8.1×10^{-5}.

All uses below occur strictly beyond these initial indices.

For $8 \leqslant P \leqslant 11$, every cost $d \leqslant L-2$ has

$$N \leqslant F_{2L-1} < \frac{2}{5} F_{2L+1} < C_{11} F_{2L+1} \leqslant C_P F_{2L+1} \leqslant m_{P/Q}.$$

For $P = 12, 13$, the same Fibonacci bound gives, for every cost $d \leqslant L-3$,

$$N \leqslant F_{2L-3} < \frac{3}{20} F_{2L+1} < C_{13} F_{2L+1} \leqslant m_{P/Q}.$$

Only the cost $d = L-2$ in the last two rows remains. Its Fibonacci maximum lies above the source, since

$$\frac{F_{2d+3}}{F_{2d+5}} > \varphi^{-2} > \frac{3}{8} > U_{12} > U_{13}.$$

For row twelve, every non-Fibonacci path satisfies $N \leqslant E_d < F_{2d+5}/3 < C_{12} F_{2d+5} \leqslant m_{12/Q}$. Here the strict envelope comparison has a short exact proof:

$$F_{2d+5} - 3E_d = 2(F_{d+1}^2 - 2F_d^2) > 0.$$

Indeed $F_{d+1}/F_d > 3/2$ for $d \geqslant 5$, and the present range starts at $d = 109$.

For row thirteen, a one-defect path is above the source:

$$\frac{N}{F_{2d+5}} \geqslant \frac{5F_{2d-1}}{F_{2d+5}} > 5\varphi^{-6} > \frac{5}{18} > U_{13}.$$

The radical inequality uses $\varphi^6 = 9 + 4\sqrt{5} < 18$. Every path with at least two defects is below the source. Since $d \geqslant 155$, Corollary 6.5 gives

$$\frac{N}{F_{2d+5}} = \frac{N}{F_{2d+3}} \frac{F_{2d+3}}{F_{2d+5}} < \frac{71}{100} \frac{5}{13} = \frac{71}{260} < C_{13}.$$

All the comparisons made in this section between C_P, the rational constants and U_P are inequalities in $\mathbb{Q}(\sqrt{5})$, decided exactly by the integer sign test of Section 5; no decimal rounding enters the proof.

Proof of Theorem 1.1. Proposition 2.2 supplies a primitive representation of cost $L = P + Q - 2$. Proposition 4.5 proves its weak minimality for $P \leqslant 7$. For each $8 \leqslant P \leqslant 13$, Proposition 6.3 covers every reduced denominator $P < Q < Q_P$. For $Q \geqslant Q_P$, the preceding three subsections exclude every smaller cost. This proves weak minimality without a denominator bound. □

Proof of Corollary 1.2. By Theorem 1.1, every occurrence with numerator at most thirteen is weakly Euclidean-minimal, and Corollary 2.6 applies to this family. □

The proof uses complete Gaussian fibers only for the finite initial windows. In the infinite tails it excludes all shorter primitive paths directly, so no factorization of an unbounded family of Markov integers is required. These two forms of completeness are complementary.

7. What the comparison isolates

The optimization problem separates two sources of difficulty in Markov uniqueness. The occurrence cost is explicit, but a fixed label may have many primitive two-square representations. Gaussian factorization describes that entire fiber without identifying its minimum. The path comparison supplies the missing order for the thirteen rows treated here. Its content is therefore stronger than a comparison between selected Markov labels, while its conclusion remains restricted to the family actually proved.

The parity-order theorem has a scope independent of the row cutoff. It turns the normalized norm for any fixed defect count into a function whose extrema on parity boxes occur at prescribed corners, even when some gaps are infinite. The useful feature is the normalization itself: bounding the numerator and Fibonacci denominator separately loses the correlation that makes narrow source intervals accessible.

Extending the result to another row requires more than another finite list of labels. One must first separate its source interval from every relevant shorter-cost stratum, including the smaller costs in the final assembly. The cover procedure has no proved termination theorem for arbitrary source intervals. A box that remains inside such an interval is an unresolved comparison, not evidence of a Markov collision. This is the precise additional work required beyond row thirteen.

The broader question is whether the conclusion of Theorem 1.1 holds on every row.

Conjecture 7.1 (Euclidean minimality of Markov representations)**.** *Every Markov occurrence weakly minimizes the Euclidean cost over the whole primitive positive two-square fiber of its label.*

By Corollary 2.6, the conjecture implies the Frobenius–Markov uniqueness conjecture. Beyond the thirteen rows the evidence is computational. By complete factorization of the labels, every one of the 4145 occurrences with denominator at most 100, or with $P + Q \leqslant 160$, is strictly minimal: its pair is the only representation of minimum cost. Figure 6 shows by how much, up to the denominator 100. The margin can be a single unit: the label $m_{4/5} = 985 = 12^2 + 29^2 = 16^2 + 27^2$ has the occurrence pair of cost 7 and

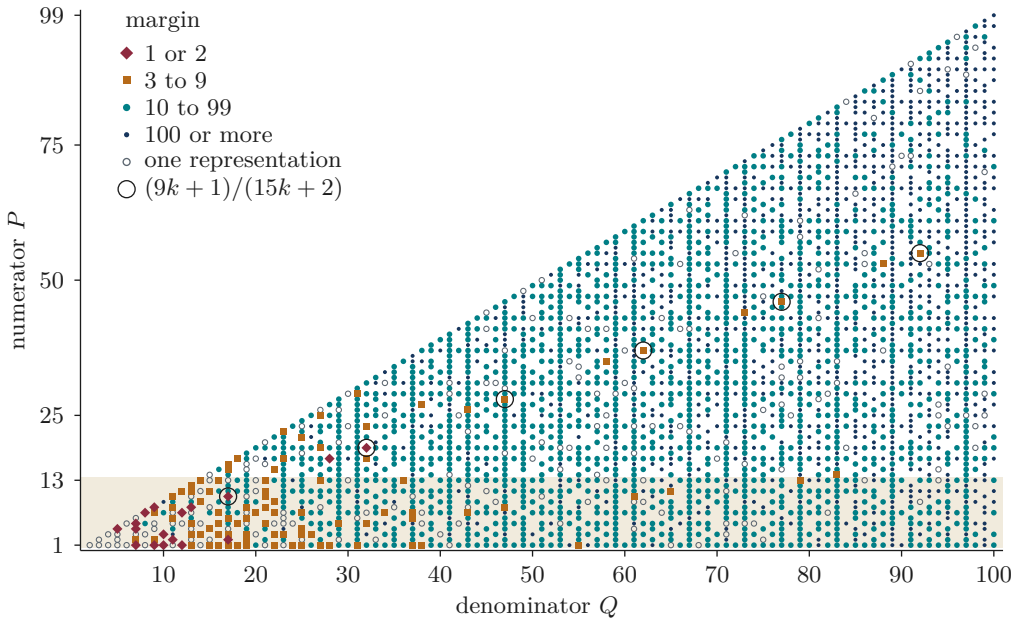


FIGURE 6. Computed evidence for Conjecture 7.1. Each mark is an occurrence P/Q with $Q \leqslant 100$, placed at its denominator and numerator; its color and shape give the margin, the cost of the cheapest other primitive representation of the label minus the occurrence cost $P+Q-2$. Hollow circles are labels with a single representation, which have no competitor. No margin is zero or negative. The small margins thin out as the denominator grows; the ringed occurrences $(9k+1)/(15k+2)$, $k = 1, \ldots, 6$, are the last ones with a margin at most k, and Proposition 7.3 explains them. The shaded rows $P \leqslant 13$ are those of Theorem 1.1.

a second pair of cost 8. Çanakçı and Schiffler checked their uniqueness conjectures by computer for all denominators below 70 [6, Remark 4.7].

7.1. **How small the margin can be.** The rarefaction of small margins in Figure 6 follows a precise law in the computed range. Among the occurrences with $P+Q \leqslant 160$, a margin at most k, for $k = 1, \ldots, 6$, occurs only when

$$P+Q \leqslant 24k+3, \qquad P \leqslant 9k+1, \qquad Q \leqslant 15k+2,$$

and all three bounds are reached by the single occurrence $(9k+1)/(15k+2)$, whose margin is exactly k (Figure 7). These six fractions are rungs of one Farey ladder, the fractions $(1+3j)/(2+5j)$ that run from $1/2$ to $3/5$, taken at $j = 3k$; their labels are the neighbors of $m_{3/5} = 433$ in the Markov tree. On every third rung the label is divisible by $5 = (1+2i)(1-2i)$, and at the six computed rungs the cheapest competitor of the occurrence pair is the element of the fiber in which the Gaussian factor above 5 is replaced by its conjugate. A companion ladder, $(2+3j)/(3+5j)$ from $2/3$ to $3/5$, behaves in the same way at $j = 3k-1$. The mechanism can be proved for every rung, and it shows that thresholds of this kind, if they exist, cannot be smaller.

The proof follows the occurrence vectors $v_j = H_j e$ of the rungs, H_j being the Gram factor of Proposition 2.2. Two rungs apart, the central words grow by one fixed word at each end, so that v_{j+2} is obtained from v_j by a matrix that depends only on the parity of j; six rungs apart, the competitor obtained by conjugating the factor above 5 is transformed by the same matrix conjugated by the rotation that multiplies by

$(1-2i)/(1+2i)$. Cost is additive under nonnegative matrices of determinant one, and the whole statement reduces to the lengths of four explicit matrices.

Lemma 7.2 (Finite facts on two ladders)**.** *For the ladder $(1+3j)/(2+5j)$ put $u = babab$ and $j_0 = 3$, and for the ladder $(2+3j)/(3+5j)$ put $u = abbab$ and $j_0 = 2$. For each ladder and each parity of $j \geqslant 1$, let $T = H_j\,\mu(u)\,H_j^{-1}$, computed at the rungs $j = 1$ and $j = 2$. At each of the four rungs j_0 and $j_0 + 3$ of the two ladders, exactly one of the two matrices R of multiplication by $(1-2i)/(1+2i)$ and by $(1+2i)/(1-2i)$ maps v_j to an integer vector, and exactly one diagonal sign matrix S makes that vector positive. Then:*

(i) *On the first ladder T is $\left(\begin{smallmatrix}1105 & 463\\ 463 & 194\end{smallmatrix}\right)$ at even j and $\left(\begin{smallmatrix}1130 & 437\\ 437 & 169\end{smallmatrix}\right)$ at odd j; on the second ladder the two matrices are exchanged.*

(ii) *At each of the four rungs, with T the matrix of its parity, the matrix $N = SRT^3R^{-1}S^{-1}$ is a nonnegative integer matrix of determinant one whose expression as a word in L_a and L_b has 50 letters, while T^3 has 48.*

(iii) *At each of the four rungs the vector SRv_j is primitive, and its Euclidean cost exceeds that of v_j by 1 at j_0 and by 2 at $j_0 + 3$.*

Proof. These are finitely many assertions about explicit integer matrices and vectors. They were verified by computer in exact arithmetic, and the verification was carried out twice independently: once through the Gram factors and the matrices of (ii), and once by Gaussian division of the occurrence pairs, with the labels taken as continuants of the continued fraction $[2, \sigma(z), 2]$. □

Proposition 7.3 (Margins along two ladders)**.** *For every integer $k \geqslant 1$, the labels of the reduced fractions $(9k+1)/(15k+2)$ and $(9k-1)/(15k-2)$ are divisible by 5. In the occurrence pair of each, written as a Gaussian integer, replacing the factor above 5 by its conjugate gives a primitive representation of the same label whose Euclidean cost exceeds the occurrence cost by exactly k. Consequently the margins of these two occurrences are at most k.*

Proof. The words. Two fractions $a/b < c/d$ with $bc - ad = 1$ are Farey neighbors, and the lower Christoffel word of their mediant is the product $\gamma_{a/b}\gamma_{c/d}$ [18]. The fraction $(1+3j)/(2+5j)$ is a neighbor of $3/5$ lying below it, since $3(2+5j)-5(1+3j) = 1$, and its mediant with $3/5$ is the next rung; starting from $\gamma_{1/2} = ab$ and $\gamma_{3/5} = ababb$, induction gives $\gamma_{(1+3j)/(2+5j)} = ab\,(ababb)^j$, with central word $z_j = b\,(ababb)^{j-1}abab$. Concatenation gives $z_{j+2} = babab \cdot z_j \cdot babab = u\,z_j\,u$. In the same way $5(2+3j) - 3(3+5j) = 1$ places $(2+3j)/(3+5j)$ above $3/5$, the words are $(ababb)^j abb$ with central words $z_j = babb\,(ababb)^{j-1}ab$, and $z_{j+2} = babba \cdot z_j \cdot abbab = \widetilde{u}\,z_j\,u$.

The transfer. In both cases the right half of the central word gains the word u at its far end while its middle letter is unchanged, so that the Gram factor of Proposition 2.2 satisfies $H_{j+2} = H_j\,\mu(u)$. Put $T_j = H_j\,\mu(u)\,H_j^{-1}$. Then $T_j v_j = H_j\,\mu(u)\,e = H_{j+2}e = v_{j+2}$, and $T_{j+2} = H_j\,\mu(u)\,\mu(u)\,\mu(u)^{-1}H_j^{-1} = T_j$, so that T_j depends only on the parity of j: it is the matrix T of Lemma 7.2(i), and $v_{j+6} = T^3 v_j$.

The competitor. Write $v_j = (s, r)^{\mathsf{T}}$, so that the label is $m = s^2 + r^2$. Fix one of the four rungs j_0, $j_0 + 3$ of Lemma 7.2, with its matrices R, S and T, and put $c_j = SR\,v_j$ for every rung j congruent to it modulo six. Since $v_{j+6} = T^3 v_j$,

$$c_{j+6} = SRT^3 v_j = N\,c_j, \qquad N = SRT^3R^{-1}S^{-1}.$$

The matrix N has nonnegative integer entries and determinant one (Lemma 7.2(ii)); it maps positive vectors to positive vectors, since no row of it vanishes, and primitive integer vectors to primitive integer vectors, since its inverse is integral. By Lemma 7.2(iii) and induction on j in steps of six, c_j is therefore a positive primitive integer vector at

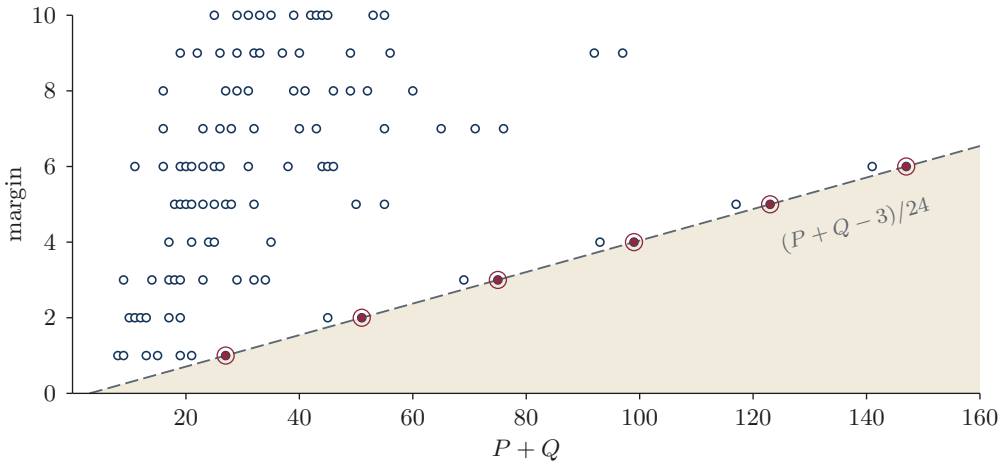


FIGURE 7. The margin against $P+Q$ for the 120 occurrences with $P+Q \leqslant 160$ and margin at most 10 (hollow circles; several occurrences may share a mark). The dashed line is $(P+Q-3)/24$, and the shaded region below it contains no occurrence. The filled wine circles are the occurrences $(9k+1)/(15k+2)$, $k=1,\dots,6$, which lie on the line and whose margin is exactly k; Proposition 7.3 shows that the margin of the next ones on the line is at most k for every k.

every rung $j=3k$ of the first ladder and $j=3k-1$ of the second. The matrices R and S preserve lengths, so c_j has norm m: up to the order of its entries it is an element of $\mathcal{F}(m)$. As a Gaussian integer, c_j is $s+ir$ multiplied by $(1\mp 2i)/(1\pm 2i)$, up to a sign and a conjugation; its integrality means that $1\pm 2i$ divides $s+ir$, so that 5 divides m, and its primitivity that 25 does not, by Proposition 3.1. The element c_j is thus the occurrence pair with the factor above 5 replaced by its conjugate.

Cost. By Lemma 2.3, every positive primitive column is $x=L_w\mathbf{1}$ for a unique word w, so that for every word w', $\ell(L_{w'}x)=\ell(L_{w'w}\mathbf{1})=|w'|+\ell(x)$. Every nonnegative integer matrix $M=\left(\begin{smallmatrix}\alpha&\beta\\ \gamma&\delta\end{smallmatrix}\right)\neq I$ of determinant one is such an $L_{w'}$. Indeed one of its rows dominates the other entrywise: if $\alpha>\gamma$ and $\beta<\delta$, then $\alpha\delta-\beta\gamma\geqslant(\gamma+1)(\beta+1)-\beta\gamma=\beta+\gamma+1$, which equals one only for $M=I$; and if $\alpha<\gamma$ and $\beta>\delta$, then $\alpha\delta-\beta\gamma\leqslant(\gamma-1)(\beta-1)-\beta\gamma<0$. Subtracting the dominated row from the dominating one removes a first letter L_a or L_b, keeps the entries nonnegative and the determinant one, and lowers the sum of the entries, so that the process ends at I. By Lemma 7.2(ii),

$$\ell(c_{j+6})=\ell(c_j)+50, \qquad \ell(v_{j+6})=\ell(v_j)+48,$$

so that the excess $\ell(c_j)-\ell(v_j)$ grows by 2 every six rungs. It is 1 and 2 at the first two rungs of each family (Lemma 7.2(iii)), hence k at the kth, which is the rung $(9k+1)/(15k+2)$ of the first ladder and $(9k-1)/(15k-2)$ of the second. The margin is at most the excess of this particular competitor. □

The proposition explains the six ringed occurrences of Figure 6: together with the computations, which find no cheaper competitor at those rungs, it gives their margins exactly. It also shows that the thresholds observed above cannot be lowered: for every k there is an occurrence with $P+Q=24k+3$ and margin at most k.

Conjecture 7.4 (Thresholds of the margin)**.** *For every integer $k\geqslant 0$, every occurrence P/Q with $P+Q>24k+3$ has margin larger than k, the margin of a label with a single primitive representation being taken as infinite.*

The case $k = 0$ asserts that every occurrence is strictly minimal, which is stronger than Conjecture 7.1 and therefore implies the Frobenius–Markov uniqueness conjecture. The conjecture holds for every occurrence with $P + Q \leqslant 160$, and Proposition 7.3 shows that it would be false with any smaller threshold.

The minimality is not a consequence of the digit pattern alone. The representation of a Markov label by its occurrence pair has a half continued fraction with digits 1 and 2 only [6, Corollary 4.4], but such digits do not force minimality for other integers. The integer $135328989953 = 37 \cdot 113 \cdot 761 \cdot 42533$, which is not a Markov number, has eight primitive representations; two of them, $196223^2 + 311168^2$ and $131468^2 + 343577^2$, have continued fractions with digits 1 and 2 only, and their costs are 28 and 27. The first is therefore not minimal. Whatever makes the conjecture true must use the Markov structure of the label, as the path comparison of this paper does on thirteen rows.

7.2. **Directives of the Christoffel words of a given length.** The comparison of this paper fixes a label and varies its representation as a sum of two squares. One may instead fix an integer m and vary the Christoffel word of length m. For each n prime to m with $1 \leqslant n < m/2$, the lower Christoffel word $\gamma_{n/m}$ has a central word, and the central word has a directive, in the sense of the iterated palindromic closure of de Luca. This subsection shows that these directives are the Euclidean directives of Lemma 2.3, characterizes those that are images of the Thue–Morse morphism, and states a conjecture on them that implies the uniqueness conjecture. It also recasts Conjecture 7.1 in terms of the square roots of -1 modulo the label.

The palindromic closure $w^{(+)}$ of a word w is the shortest palindrome having w as a prefix, and the iterated palindromic closure is defined by $\mathrm{Pal}(\varepsilon) = \varepsilon$ and $\mathrm{Pal}(vx) = (\mathrm{Pal}(v)x)^{(+)}$ for a word v and a letter x [7]. The map Pal is injective, and d is called the *directive* of the central word $\mathrm{Pal}(d)$; the map $d \mapsto a\,\mathrm{Pal}(d)b$ is a bijection from the binary words onto the proper lower Christoffel words [19, Section 2 and the proof of Theorem 1], a consequence of [3, Corollary 3.1]. The Thue–Morse morphism θ sends a to ab and b to ba; its image is the set $\{ab, ba\}^*$ of the products of the words ab and ba.

Lemma 7.5 (Directives of Christoffel words)**.** *Let $1 \leqslant n < m/2$ be coprime integers, write $\gamma_{n/m} = azb$, and let d be the directive of z. Then $L_d\mathbf{1} = (m - n, n)^{\mathsf{T}}$: the word d is the directive of the pair $(m - n, n)$ in the sense of Lemma 2.3, and $|d| = \ell(n, m) - 1$. If $n = 1$, then $d = a^{m-2}$. If $n \geqslant 2$ and $m/n = [c_0; c_1, \ldots, c_k]$ is the canonical continued fraction, the runs of d have the lengths $c_0 - 1, c_1, \ldots, c_{k-1}, c_k - 1$, alternately of the letters a and b, beginning with a. In particular, d lies in $\{ab, ba\}^*$ if and only if*

$$\frac{m}{n} = [2; B_1, \ldots, B_j, 2], \qquad j \geqslant 0, \quad B_i \in \{(2), (1,1)\}, \tag{39}$$

that is, if and only if the first and the last partial quotients equal 2 and those in between split into consecutive blocks (2) and $(1, 1)$.

Proof. The pair. Berthé, de Luca and Reutenauer attach to a word d the incidence matrix M_d of a composition of morphisms, with $M_{dd'} = M_d M_{d'}$, $M_a = \left(\begin{smallmatrix} 1 & 1 \\ 0 & 1 \end{smallmatrix}\right) = L_a$ and $M_b = \left(\begin{smallmatrix} 1 & 0 \\ 1 & 1 \end{smallmatrix}\right) = L_b$ [3, proofs of Proposition 3.1 and Corollary 3.2], so that $M_d = L_d$. By [3, Corollary 3.1], the Christoffel word $a\,\mathrm{Pal}(d)b$ has as many letters a as the sum of the entries of the first row of L_d, and as many letters b as the sum of the entries of its second row; that is, its two letter counts are the entries of $L_d\mathbf{1}$. For $z = \mathrm{Pal}(d)$, (6) therefore gives $L_d\mathbf{1} = (m - n, n)^{\mathsf{T}}$, and Lemma 2.3 shows that d is the directive of $(m - n, n)$, of length $\ell(m - n, n)$. Since $m - n > n$, the first subtraction carries the pair (n, m) to $(n, m - n)$, so that $\ell(n, m) = 1 + \ell(n, m - n)$.

The runs. If $n = 1$, the pair $(m - 1, 1)$ reaches $(1, 1)$ after $m - 2$ subtractions of its second coordinate, and its directive is a^{m-2}. If $n \geqslant 2$, then m/n is not an integer, so

$k \geqslant 1$; moreover $c_0 \geqslant 2$ because $m > 2n$, and $(m-n)/n = m/n - 1 = [c_0 - 1; c_1, \dots, c_k]$ is a canonical expansion. The description of directives after Lemma 2.3 gives the runs $c_0 - 1, c_1, \dots, c_{k-1}, c_k - 1$, alternately of a and b, beginning with a.

The image of θ. A word lies in $\{ab, ba\}^*$ exactly when its length is even and its letters in the positions $2i-1$ and $2i$ differ for every i, that is, when a run ends after every odd position. If $n = 1$, the word a^{m-2}, with $m \geqslant 3$, has no such run end, and (39) fails since $m/1$ has a single partial quotient. Let $n \geqslant 2$, and write $r_0 = c_0 - 1$, $r_i = c_i$ for $0 < i < k$ and $r_k = c_k - 1$, all positive; the runs end after the positions r_0, $r_0 + r_1$, …, $r_0 + \cdots + r_{k-1}$. Suppose that d lies in $\{ab, ba\}^*$, of length $2h$. A run ending after the position 1 forces $r_0 = 1$, that is $c_0 = 2$. A run ending after the last odd position $2h-1$ leaves a last run of length 1, so $r_k = 1$, that is $c_k = 2$. Between two consecutive odd positions $2i-1$ and $2i+1$, there is either no other run end, which makes a run of length 2, or one run end after the position $2i$, which makes two runs of length 1; so the runs $r_1, \dots, r_{k-1}$, which are the quotients $c_1, \dots, c_{k-1}$, split into consecutive blocks (2) and $(1,1)$. Conversely, when (39) holds, the run lengths are 1, then blocks (2) and $(1,1)$, then 1, so the word has even length and a run ends after every odd position. □

Every occurrence supplies an integer of the form (39). Let P/Q be an occurrence of the label m, with central word z, and write $\mu(z)e = (x, y)^{\mathsf{T}}$. Since $\mu(a) = P_1^2$, $\mu(b) = P_2^2$ and $P_2(1,0)^{\mathsf{T}} = e$, the column $P_2\,\mu(z)\,P_2(1,0)^{\mathsf{T}} = P_2\,\mu(z)\,e = (2x+y, x)^{\mathsf{T}}$ is the image of $(1,0)^{\mathsf{T}}$ under the product of the matrices P_c, c running through the sequence $(2, \sigma(z), 2)$ of Section 2, and its first entry $2x + y = e^{\mathsf{T}}\mu(z)e$ is m by (8). By Lemma 2.1, the integer $u_{P/Q} = x$, which we call the *root* of the occurrence, satisfies

$$\frac{m}{u_{P/Q}} = [2; \sigma(z), 2]. \tag{40}$$

The sequence $\sigma(z)$ consists of the blocks $(1,1)$ and $(2,2)$, so the root is of the form (39), and $m/3 < u_{P/Q} < m/2$. The matrix $P_2\,\mu(z)\,P_2$ is symmetric of determinant one, with first column $(m, u_{P/Q})^{\mathsf{T}}$, so that $u_{P/Q}^2 \equiv -1 \pmod m$. Since $\mu(a)e_1 = e$, $\mu(b)e_1 = 2e + e_1$ and $\mu(z)$ is symmetric, the upper left entry of $\mu(\gamma_{P/Q}) = \mu(a)\mu(z)\mu(b)$ is $e^{\mathsf{T}}\mu(z)(2e + e_1) = 2m + u_{P/Q}$. By the first step of the proof of Proposition 3.4, $u_{P/Q}\,m_R \equiv -m_L \pmod m$, and since $0 < u_{P/Q} < m/2$, the root is the characteristic number of the Markov triple (m, m_L, m_R) in the normalization of [1, Section 2.2], the smaller of the two solutions in $(0, m)$ of $m_R x \equiv \pm m_L \pmod m$.

Conjecture 7.6 (Directives in the image of the Thue–Morse morphism)**.** *For every Markov number $m \geqslant 5$, exactly one integer n with $1 \leqslant n < m/2$, prime to m, has the property that the directive of the central word of $\gamma_{n/m}$ lies in $\{ab, ba\}^*$; equivalently, by Lemma 7.5, exactly one such n satisfies* (39)*.*

Proposition 7.7 (The conjecture implies uniqueness)**.** *Conjecture 7.6 implies the Frobenius–Markov uniqueness conjecture. In terms of the Thue–Morse morphism, it states that for every Markov number $m \geqslant 5$ exactly one word w beginning with the letter a satisfies $|\mathrm{Pal}(\theta(w))| = m - 2$.*

Proof. Uniqueness. Every occurrence P/Q of a Markov number m supplies the integer $n = u_{P/Q}$, which satisfies (39) by (40). Two distinct occurrences supply distinct integers. Indeed, the canonical continued fraction (40) of $m/u_{P/Q}$ determines the sequence $\sigma(z)$; this sequence determines z, because σ replaces each letter by two equal entries that identify it; and z determines the word $\gamma_{P/Q} = azb$ and the fraction P/Q, by (6). If the integer of the conjecture is unique, so is the occurrence.

The second form. Let n be an integer as in the conjecture, and d the directive of the central word of $\gamma_{n/m}$. By Lemma 7.5, d begins with a, and it lies in $\{ab, ba\}^*$ exactly

when $d = \theta(w)$ for a word w, which then begins with a and is unique, the morphism θ being injective. Conversely, let w begin with a. Then $d = \theta(w)$ begins with a, and $a\,\mathrm{Pal}(d)b$ is a proper lower Christoffel word whose numbers of letters a and b are the entries of $L_d\mathbf{1}$, by the first step of the proof of Lemma 7.5; the first entry is the larger, since $L_d\mathbf{1} = L_aL_{d'}\mathbf{1}$ with $d = ad'$. This word is therefore $\gamma_{n/m}$ for coprime integers with $n < m/2$ and $m = |\mathrm{Pal}(\theta(w))| + 2$, and the directive of its central word is $\theta(w)$. The integers n of the conjecture for the label m thus correspond one to one to the words w beginning with a such that $|\mathrm{Pal}(\theta(w))| = m - 2$. □

Reutenauer and Vuillon proved that, for every word v, the number $|\mathrm{Pal}(\theta(\mathrm{Pal}(av)))|+2$ is a Markov number, that every Markov number $m \geqslant 5$ arises in this way, and that the injectivity of the map $v \mapsto |\mathrm{Pal}(\theta(\mathrm{Pal}(av)))|$ is equivalent to the uniqueness conjecture [19, Theorem 1]. The words $\mathrm{Pal}(av)$ are the central words beginning with a. In the second form of Proposition 7.7, the conjecture contains this injectivity, and asserts in addition that no word w beginning with a that is not a central word gives a Markov number. The next lemma supports it for every Markov number up to 10^{12}.

Lemma 7.8 (Finite facts on directives)**.** (i) *There are* 150 *Markov numbers m with $5 \leqslant m \leqslant 10^{12}$. Each is the label of exactly one occurrence P/Q, and its root $u_{P/Q}$ is the only integer n prime to m with $1 \leqslant n < m/2$ satisfying* (39).

(ii) *The integers* 31 *and* 1130*, which are not Markov numbers, have exactly two such integers prime to them each:* $31/12 = [2; 1, 1, 2, 2]$ *and* $31/13 = [2; 2, 1, 1, 2]$*, and* $1130/437 = [2; \sigma(abba), 2]$ *and* $1130/467 = [2; \sigma(baab), 2]$.

(iii) *For $m = m_{9/10} = 6\,625\,109$, the root is $u_{9/10} = 2\,744\,210$, with $m/u_{9/10} = [2; 2, \ldots, 2]$, eighteen partial quotients equal to* 2*, so that $\ell(u_{9/10}, m) = 35$. The integer $n = 2\,561\,343$ is prime to m and smaller than $m/2$, and*

$$\frac{m}{n} = [2; 1, 1, 2, 2, 1, 1, 2, 1, 1, 1, 2, 1, 1, 1, 2, 1, 2, 2, 1, 1, 1, 1, 2, 2],$$

so that $\ell(n, m) = 34$.

Proof. These are finitely many assertions about integers. They were verified by computer in exact arithmetic, and the verification was carried out twice independently. The first verification generates the labels from the Markov triples of Lemma 3.5, takes the roots from the residues of $-m_L/m_R$ modulo m given by the first step of the proof of Proposition 3.4, and searches, for each label, the sequences of blocks from the left. If a prefix $[2; B_1, \ldots, B_i]$ has p and p' as the numerators of its last two convergents, every completion has a tail column $(X, Y)^{\mathsf{T}}$ with coprime entries $0 < Y < X$ and $m = pX + p'Y$; hence $Y < m/(p + p')$ and Y lies in one residue class modulo p, so that once $p(p + p') > m$ at most one candidate remains, and it is tested directly. The second verification enumerates all the continued fractions (39) with numerator at most 10^{12} and compares their numerators and denominators with the labels and roots computed from the Cohn matrices (7). The continued fractions of the last two items can also be checked by hand. □

The integers 31 and 1130 of Lemma 7.8 show that Conjecture 7.6 depends on the Markov structure of m. Whenever the block sequence $B_1, \ldots, B_j$ of (39) is not a palindrome, the reversed expansion has the same numerator, since the product of the symmetric matrices P_c in reverse order is the transpose, and a different denominator; this is the case of 31. For 1130, the two expansions come from the palindromes *abba* and *baab*, which are not central words and have the same label $e^{\mathsf{T}}\mu(z)e$. The conjecture therefore asserts that a Markov number is the numerator of no expansion (39) with a non-palindromic block sequence, and of no palindromic one other than that of its root.

The roots of the occurrences are square roots of -1, and the square roots of -1 carry the whole two-square fiber, as the next lemma shows. This gives Conjecture 7.1 a form in which only the label and its square roots of -1 appear. In substance the lemma is classical. Smith derived from two identities of Euler that the numerator of a palindromic continued fraction with an even number of partial quotients is a sum of two coprime squares, given by its first half. He proved in this way that every prime $p \equiv 1 \pmod 4$ is a sum of two squares, since a parity argument shows that one of the expansions of p/u with $1 < u < p/2$ is a palindrome, and he observed that the same argument applies to every divisor of an integer $1+x^2$ [21]. Brillhart read the two squares off the remainders of the Euclidean algorithm on p and the square root u of -1, refining a method of Hermite [4]. Çanakçı and Schiffler relate the primitive representations of an integer to its even palindromic continued fractions [6, Theorem 3.10 and Corollary 3.14]. What the lemma adds is a complete proof, for every label, of the correspondence between the square roots of -1 in $(1, m/2)$ and the fiber $\mathcal{F}(m)$, and the identity (41) between the costs, on which the reformulation of the conjecture below rests.

Lemma 7.9 (Square roots of -1)**.** *Let $m \geqslant 5$ be an integer with a primitive representation as a sum of two squares, and let u be an integer with $1 < u < m/2$ and $u^2 \equiv -1 \pmod m$. Then the canonical continued fraction of m/u is a palindrome $[c_0; c_1, \dots, c_{2h-1}]$ of even length. If $s/r = [c_{h-1}; c_{h-2}, \dots, c_0]$ in lowest terms, then $m = r^2 + s^2$, $0 < r < s$, and*

$$\ell(u, m) = 2\,\ell(r, s) + 1. \tag{41}$$

The map $u \mapsto (r, s)$ is a bijection from these integers u onto $\mathcal{F}(m)$, and it sends the root of an occurrence to its occurrence pair.

Proof. The palindrome. Write $m/u = [c_0; c_1, \dots, c_k]$ canonically. Here $k \geqslant 1$, because $u > 1$ is prime to m; $c_0 \geqslant 2$, because $u < m/2$; and $c_k \geqslant 2$. By Lemma 2.1, the product $\Pi = P_{c_0} \cdots P_{c_k}$ has first column $(m, u)^{\mathsf{T}}$; write $\Pi = \left(\begin{smallmatrix} m & m' \\ u & u' \end{smallmatrix}\right)$. Each P_c is symmetric, so $\Pi^{\mathsf{T}} = P_{c_k} \cdots P_{c_0}$, whose first column is $(m, m')^{\mathsf{T}}$, and Lemma 2.1 gives $m/m' = [c_k; \dots, c_0]$, a canonical expansion since $c_0 \geqslant 2$. As $c_k \geqslant 2$ and $k \geqslant 1$, $m/m' > 2$, so that $0 < m' < m/2$. The determinant of Π is $(-1)^{k+1}$, so $mu' - m'u = (-1)^{k+1}$, and $m'u \equiv (-1)^k \pmod m$. Since $u^2 \equiv -1$, the inverse of u modulo m is $-u$, whence $m' \equiv (-1)^{k+1}u \pmod m$. If k were even, then $m' \equiv -u$ and $0 < m' < m$ would give $m' = m - u > m/2$, a contradiction. Hence k is odd, $m' \equiv u$, and $m' = u$, both lying in $(0, m)$. The two canonical expansions $[c_0; \dots, c_k]$ and $[c_k; \dots, c_0]$ of m/u then coincide, so that $c_i = c_{k-i}$ for every i, and $k + 1 = 2h$ is even.

The two squares. By the symmetry of the expansion, the second half of the product is $P_{c_{h-1}} \cdots P_{c_0} = \Pi_0^{\mathsf{T}}$, where $\Pi_0 = P_{c_0} \cdots P_{c_{h-1}}$, so that $\Pi = \Pi_0 \Pi_0^{\mathsf{T}}$. If (s', r') is the first row of Π_0, the upper left entry of Π is $m = s'^2 + r'^2$. The first column of $\Pi_0^{\mathsf{T}} = P_{c_{h-1}} \cdots P_{c_0}$ is $(s', r')^{\mathsf{T}}$, so Lemma 2.1 shows that s' and r' are coprime and positive, with $s'/r' = [c_{h-1}; \dots, c_0]$, a canonical expansion since $c_0 \geqslant 2$; hence $(s', r') = (s, r)$, and $r < s$ because this expansion is not [1]. Finally (4) gives $\ell(u, m) = 2(c_0 + \cdots + c_{h-1}) - 1$ and $\ell(r, s) = c_0 + \cdots + c_{h-1} - 1$, which is (41).

The bijection. The pair (r, s) determines u, because the canonical expansion of s/r is the reversed first half of the palindrome, which gives the whole palindrome and hence m/u. The map is therefore injective. By Proposition 3.1, $|\mathcal{F}(m)| = 2^{t-1}$, where t is the number of distinct odd primes dividing m, as in Proposition 3.1. By the Chinese remainder theorem and Hensel's lemma, -1 has exactly 2^t square roots modulo m, two modulo each odd prime power dividing m and one modulo 2 when m is even. They pair off as u and $m - u$, and none of them is 1 or $m/2$, because $m \geqslant 5$; so exactly 2^{t-1} of them lie in $(1, m/2)$. An injective map between two finite sets of the same size is a bijection. Finally, for the root of an occurrence, the expansion is (40), a palindrome

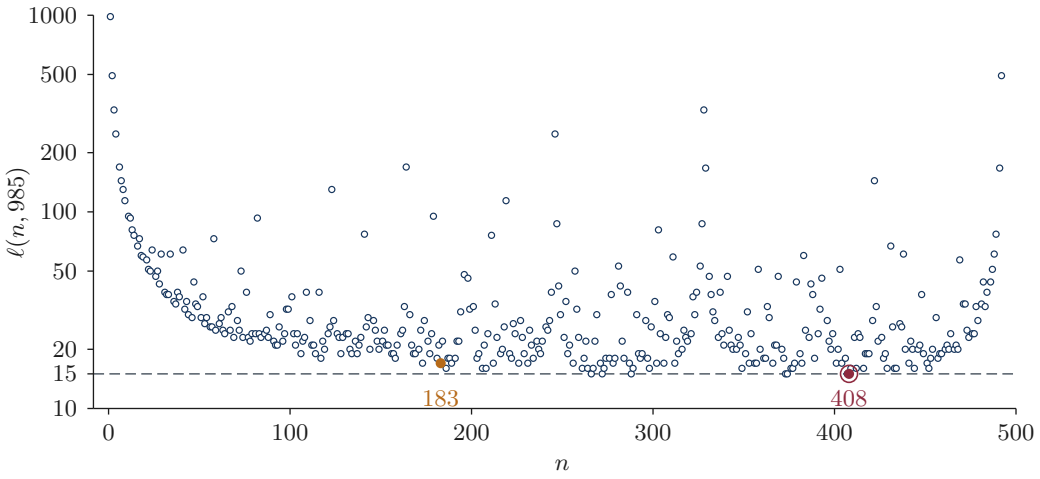


FIGURE 8. The Euclidean cost $\ell(n, m)$ for $m = m_{4/5} = 985$ and the 392 integers n prime to m with $1 \leqslant n < m/2$ (hollow circles), on a logarithmic scale; the directive of the central word of $\gamma_{n/m}$ has $\ell(n, m) - 1$ letters. The two square roots of -1 modulo m in $(1, m/2)$ are filled: the root 408 of the occurrence $4/5$, of cost $15 = 2 \cdot 7 + 1$, and 183, of cost $17 = 2 \cdot 8 + 1$, which correspond by Lemma 7.9 to the representations $12^2 + 29^2$ and $16^2 + 27^2$. Six other integers, 266, 272, 288, 373, 374 and 407, share the least cost 15 with the root (dashed line), but only the root, ringed, satisfies (39).

because z is one, and the reversal of its first half is the right half $[\sigma(w), 2]$ or $[j, \sigma(w), 2]$ of the paragraph after Proposition 2.2, which is the continued fraction of the occurrence pair. □

By Lemma 7.9 and Proposition 2.2, Conjecture 7.1 is equivalent to the following statement: for every occurrence P/Q of a label m, no square root of -1 modulo m in $(1, m/2)$ has a shorter Euclidean algorithm with m than the root, whose cost is $\ell(u_{P/Q}, m) = 2(P + Q - 2) + 1$. Figure 8 shows the label $m_{4/5} = 985$, whose two representations differ in cost by one. The restriction to square roots of -1 cannot be dropped, even on the rows of Theorem 1.1: by the last item of Lemma 7.8, the integer $n = 2\,561\,343$, which is not a square root of -1 modulo $m_{9/10}$ since its cost is even, has a shorter Euclidean algorithm with $m_{9/10}$ than the root. Among all the integers n prime to m, the condition (39) isolates the root where the minimum of the cost does not.

Conjecture 7.6 is a rule of the kind discussed in Remark 3.9: it designates the root from the label alone, and with it the occurrence pair, which Lemma 7.9 reads off the first half of the continued fraction of $m/u_{P/Q}$. Like Conjecture 7.1, it can hold only because of the Markov structure of the label, as the integers 31 and 1130 of Lemma 7.8 show, and it reaches the uniqueness conjecture by a different road: through the palindromic closure and the Thue–Morse morphism rather than through the comparison of costs.

SUPPLEMENTARY MATERIAL

The supplementary material accompanying this article contains the exact computations behind Proposition 6.1 and Lemma 7.2: the factorizations and complete Gaussian fibers of the finite windows, the covers of the unbounded tails, the source comparisons, the matrices of the two ladders, two independent verifications of each, and instructions for repeating them. It also contains the complete factorizations of the labels behind the

margins of Section 7, verified a second time independently, and the two independent computations behind Lemma 7.8.

Code and data availability

The source code and the data supporting this article, with the instructions that repeat every calculation, are contained in the supplementary material of this article, which is available, together with that of the other articles of this series, at

https://github.com/dutykh/fibonacci-pell-nearest-gaps/

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Generative AI (Claude Code by Anthropic) was used exclusively for grammatical polishing, LaTeX typesetting, and generating TikZ figure templates. No AI tools were used in the formulation, derivation, or verification of the mathematical results.

Conflict of interest

The authors declare that they have no conflict of interest.

References

1. Martin Aigner, *Markov's theorem and 100 years of the uniqueness conjecture: A mathematical journey from irrational numbers to perfect matchings*, Springer International Publishing, 2013.
2. Arthur Baragar, *On the unicity conjecture for Markoff numbers*, Canadian Mathematical Bulletin **39** (1996), no. 1, 3–9.
3. Valérie Berthé, Aldo de Luca, and Christophe Reutenauer, *On an involution of Christoffel words and Sturmian morphisms*, European Journal of Combinatorics **29** (2008), 535–553.
4. John Brillhart, *Note on representing a prime as a sum of two squares*, Mathematics of Computation **26** (1972), no. 120, 1011–1013.
5. J. O. Button, *The uniqueness of the prime Markoff numbers*, Journal of the London Mathematical Society (2) **58** (1998), no. 1, 9–17.
6. İlke Çanakçı and Ralf Schiffler, *Snake graphs and continued fractions*, European Journal of Combinatorics **86** (2020), Article 103081.
7. Aldo de Luca, *Sturmian words: Structure, combinatorics, and their arithmetics*, Theoretical Computer Science **183** (1997), no. 1, 45–82.
8. Denys Dutykh and Laurent Vuillon, *Endpoint costs and balanced near-minimizers under Gaussian reflections*, 2026, Submitted; full text in the repository given at the end of this article.
9. ———, *Rational reflections and palindromic words in the Cohn monoid*, 2026, Submitted; full text in the repository given at the end of this article.
10. Ferdinand Georg Frobenius, *Über die Markoffschen Zahlen*, Sitzungsberichte der Königlich Preussischen Akademie der Wissenschaften zu Berlin (1913), 458–487.
11. Jonah Gaster, *Boundary slopes for the Markov ordering on relatively prime pairs*, Advances in Mathematics **403** (2022), Article 108377.
12. I. D. Kan, *Methods for estimating continuants*, Journal of Mathematical Sciences **182** (2012), no. 4, 508–517.
13. ———, *Methods for estimating of continuants, corrected version*, 2021, arXiv:2106.03789v1.
14. Clément Lagisquet, Edita Pelantová, Sébastien Tavenas, and Laurent Vuillon, *On the Markov numbers: Fixed numerator, denominator, and sum conjectures*, Advances in Applied Mathematics **130** (2021), Article 102227.
15. Mong Lung Lang and Ser Peow Tan, *A simple proof of the Markoff conjecture for prime powers*, Geometriae Dedicata **129** (2007), 15–22.

16. Kyungyong Lee, Li Li, Michelle Rabideau, and Ralf Schiffler, *On the ordering of the Markov numbers*, Advances in Applied Mathematics **143** (2023), Article 102453.
17. Michelle Rabideau and Ralf Schiffler, *Continued fractions and orderings on the Markov numbers*, Advances in Mathematics **370** (2020), Article 107231.
18. Christophe Reutenauer, *Christoffel words and Markoff triples*, Integers **9** (2009), no. 3, 327–332.
19. Christophe Reutenauer and Laurent Vuillon, *Palindromic closures and Thue–Morse substitution for Markoff numbers*, Uniform Distribution Theory **12** (2017), no. 2, 25–35.
20. Paul Schmutz, *Systoles of arithmetic surfaces and the Markoff spectrum*, Mathematische Annalen **305** (1996), no. 1, 191–203.
21. Henry John Stephen Smith, *De compositione numerorum primorum formae* $4\lambda + 1$ *ex duobus quadratis*, Journal für die reine und angewandte Mathematik **50** (1855), 91–92.
22. Ying Zhang, *Congruence and uniqueness of certain Markoff numbers*, Acta Arithmetica **128** (2007), no. 3, 295–301.

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